TECHNICAL MANUAL & GUIDE

Student's Lab Workbook: A companion to Python for Introductory Statistics

Companion exercises for statistical analysis using Python and Google Colaboratory

Questions, comments or errors can be reported by sending an email to saman2@ccc.edu. A link to an interactive Colab version of this document can be found in this interactive Colab workbook.

Keyboard Shortcut for HTML (Weblink) and PDF Users: When you click an external link and want to return to where you were before the click, use `Alt` + `Left Arrow` on Windows and `Command` + `Left Arrow` on Mac.

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1. Introduction to Python

Python is a general-purpose simple-syntax programming language created by Guido van Rossum in 1989. It is used in scientific computing, data science, web development, software development, artificial intelligence, machine learning etc.

1.1. Getting Started

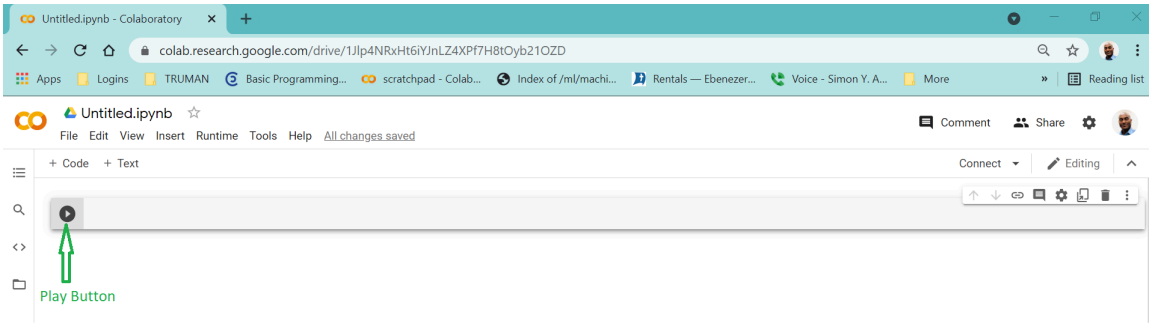
Google Colaboratory Notebook (**Colab** for short) is a free web-based Google product, similar to Google Docs, that requires no software installation for Python programming. A Colab Notebook is a list of **cells** created by adding a **code** or a **text** cell. When you create your own Colab notebooks, they are stored in your Google Drive account.

To Open a New Colab Notebook

Watch demo video

1. Click this link.
2. Sign in with your Google account. If you are already signed in, continue to the next step. You can create a free Google account.
3. Click **New Notebook** from the bottom-right of the pop-up window. Note: It may open a new notebook automatically sometimes.

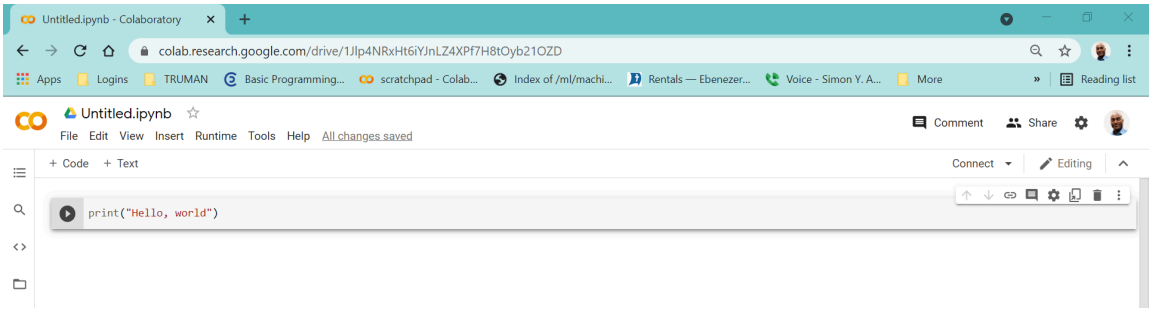
Note: Google Colab Notebook (Picture Below)



To Write Your First Python Code

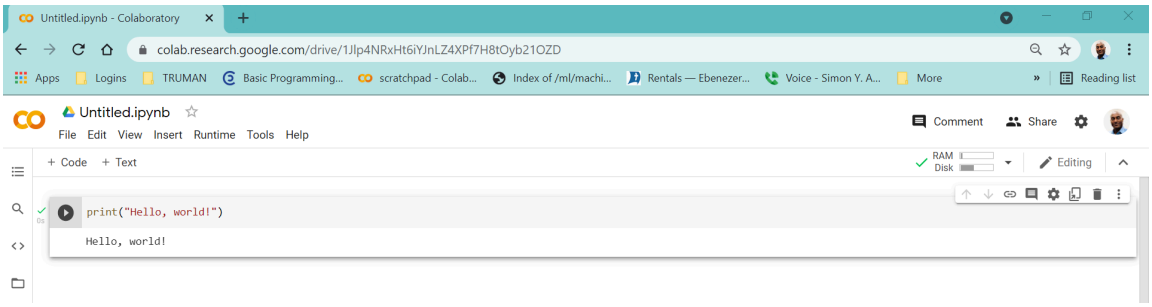
1. Click the code-cell next to the `play` button
2. Type `print("Hello, World!")`

See picture below



To Execute(run) Your First Python Code

1. Click the `play` button located before `print("Hello, World!")`
2. Or click on your first Python code and press Ctrl and Enter on your keyboard. Command and Enter for Mac(Apple) computers.
3. `Hello, world!` is displayed on a new line

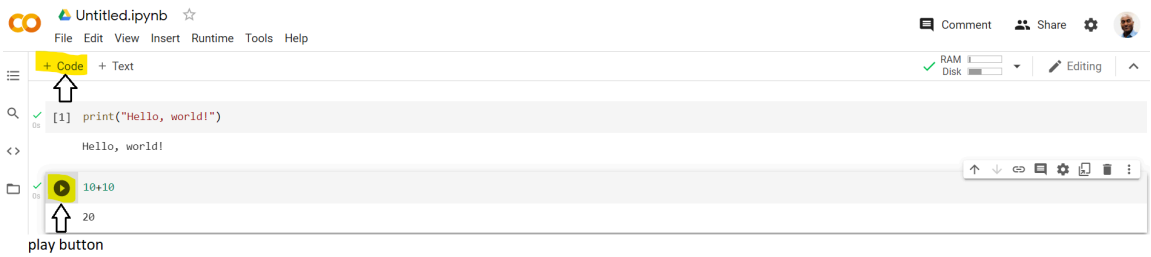


To Edit the Code

1. Click the code-cell with `print("Hello, World!)"`
2. Start editing

To Write Your Second Python Code

1. Click `+ Code` in your notebook
2. Type `10+10` in the new code cell
3. Click the `play` button
4. The output `20` is displayed on a new line



play button

Note: If you prefer **not** to use Google Colab Notebook, you can install free Python in your computer. See appendix 1 for details.

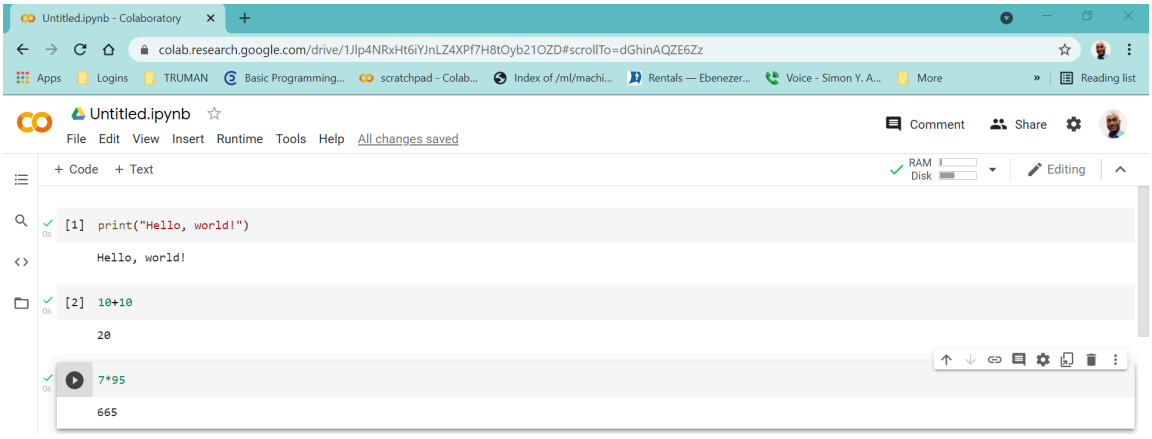
Python Arithmetic Operators

Name	Operator	Example	Result
Addition	`+`	`5+6`	`11`
Subtraction	`-`	5-6	`-1`
Multiplication	`*`	`5*6`	`30`
Division	`/`	`5/6`	`0.83333`
Exponentiation	`**`	`5**2`	5^2 = `25`

To Write Your Third Python Code

7×95

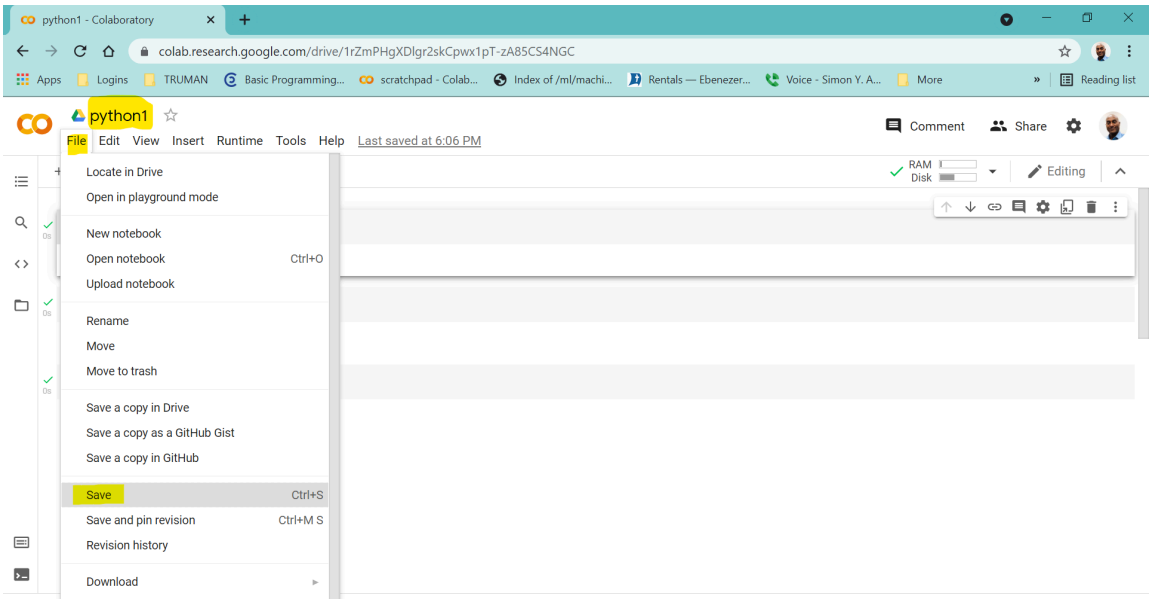
1. Click `+ Code` on a notebook to add a code cell
2. Type `7*95`
3. Press the `play` button to execute the code
4. `665` is the result



To Save Your Colab Notebook

Watch demo video

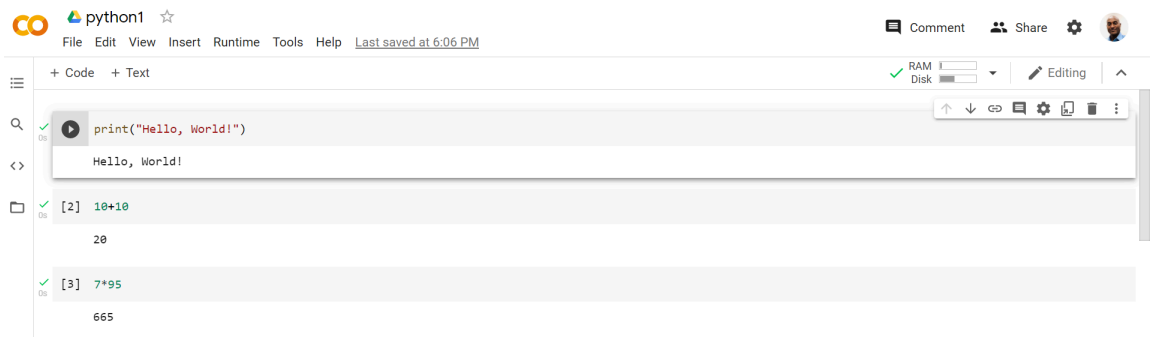
1. Click `Untitled.ipynb` at the top of the notebook
2. Delete the title and replace it by typing `python1`
3. Press the `enter` key
4. The file `python1` is now saved in your Google Drive
5. Click File and then Save
6. Close your browser



To Open Your python1 Notebook

1. Click this link or go to your Google Drive.
2. Click `python1` from the list on the pop-up window
3. The notebook will open

OR click File menu then Open Notebook



General Direction: For the lab questions below:

1. Add a new code-cell on a Colab notebook
2. Type the code based on the statement of the lab
3. Execute the code

Lab 1.1.1

Add a code-cell and type `print("Hello, Python!")` and press the `play` button.

view answer

In [ ]:
```
print("Hello, Python!")
```

Lab 1.1.2

Add a code-cell to your Colab notebook and write a Python code that displays the phrase "`Welcome to Python Programming Language.`"

view answer

In [ ]:
```

```

1.2 Python Arithmetic Operators

Name	Operator	Example	Result
Addition	+	5+6	11

Name	Operator	Example	Result
Subtraction	`-`	5-6	`-1`
Multiplication	`*`	`5*6`	`30`
Division	`/`	`5/6`	`0.83333`
Exponentiation	`**`	`5**2`	5^2 = `25`

Lab 1.2.1

Add a code-cell, type `10 + 10`, and click the `play` button to the left of `10 + 10`.

view answer

In [ ]:
```
10+10
```

Lab 1.2.2

Write a Python code to find the value of $100 - 88$.

view answer

In [ ]:
```

```

Lab 1.2.3

Write a Python code to perform the operation $\frac{1}{2} + 7.1 - 10 \times 3$. On a code-cell type `1/2 + 7.1 - 10*3` and run (execute) the code by pressing the Ctrl+Enter or click the `play` button.

view answer

In [ ]:
```

```

Lab 1.2.4

Write a Python code to perform the operation $-5 + 7 - \frac{100+0.05}{10}$.

view answer

In [ ]:
```

```

Lab 1.2.5

Write a Python code to perform the operation $\frac{1}{671000000}$.

view answer

Lab 1.2.6

For the year 2021, the Illinois state tax is 6.25%. How much state tax will one pay for a car priced at $16,000? Write a Python code to calculate the tax.

view answer

In [ ]:

Lab 1.2.7

Write a Python code to evaluate: $28/4 \times 2$.

view answer

In [ ]:

Watch demo video

Lab 1.2.8

Write a Python code to perform the operation: $7^2 + \frac{1}{\sqrt{2}}$.

view answer

In [ ]:

Lab 1.2.9

Write a code for: $7^2 - \sqrt{8}$

view answer

In [ ]:

Lab 1.2.10

Write a code to evaluate: $\frac{7^2 - \sqrt{8}}{25}$

view answer

In [ ]:

1.3 Python Variables

Watch demo video

Lab 1.3.1

Given the ordered pairs `(6,9)` and `(5,1)`, write a Python code to find the slope and midpoint.

view answer

In [ ]:

Lab 1.3.2

To solve a quadratic equation of the form $ax^2 + bx + c = 0$, we can use the formula below:

$x = \frac{-b-\sqrt{b^2-4ac}}{2a}$ or $x = \frac{-b+\sqrt{b^2-4ac}}{2a}$

Write a Python code to solve $2x^2 - 3x - 4$ using the quadratic formula.

view answer

In [ ]:

Lab 1.3.3

Evaluate $\frac{1+2}{18}$ and round the result to one decimal place.

view answer

In [ ]:

1.4 Python Comments

Lab 1.4.1

Write a Python code to find the area of a triangle with length 5 inches and width 8 inches. Use comments to explain your steps.

view answer

The Numpy Library

Watch demo video

Lab 1.5.1

Write a code to perform the operations: $|7 \times 2 - \sqrt{89}|$

view answer

In [ ]:

Lab 1.5.2

Write a code to evaluate: $\frac{|7^2-\sqrt{8}|}{25}$

view answer

In [ ]:

Lab 1.5.3

The area of a triangle with two sides of a & b and an angle between the two sides C is given by the formula $A = a \times b \times \sin(C)$. Where $\sin(C)$ is the value of the sine function at C.

Find the area of a triangle with sides 10 and 12 inches and included angle of $\frac{\pi}{3}$.

view answer

In [ ]:

Lab 1.5.4

Write a Python code to perform the operations: $\sqrt{6^2 - 4 \times 2 \times (-5)}$

view answer

1.6 `input` Data From Keyboard

Watch demo video

Lab 1.6.1

Write a Python code that asks an adult user to provide weight and height. Use those values to compute the BMI. Weight in lbs and height in foot and inches.

view answer

2. Python List

Watch demo video

2.1 Creating and Accessing Lists

Lab 2.1.1

Write a Python code to create a Python **list** containing the following items: `1,2,3,4,5,6,7,8,9`

- Display all items

view answer

Lab 2.1.2

Write a Python code to create a Python **list** contianing the following items: `1,2,3,4,5,6,7,8,9`

- Display the first item

view answer

Lab 2.1.3

Write a Python code to list the following items: `1,2,3,4,5,6,7,8,9`

- Display the last item

view answer

Lab 2.1.4

Write a Python code to create a Python **list** containing the following items: `1,2,3,4,5,6,7,8,9`

- Display from 2nd to 4th items.

view answer

2.2 List Operations

Lab 2.2.1

Create a Python list using the ages of US Supreme Course Judges (as of 2021): `72,82,66,70,66,60,53,56,49`

- write a Python code to sort the ages

view answer

Lab 2.2.2

Create a Python list using the ages of US Supreme Course Judges (as of 2021): `72,82,66,70,66,60,53,56,49`

- write a Python code to display the oldest age

view answer

Lab 2.2.3

Create a Python list using the ages of US Supreme Course Judges (as of 2021): `72,82,66,70,66,60,53,56,49`

- write a Python code to display a list from the oldest to the youngest

view answer

3. Python Syntax

3.1 `if` statements

Watch demo video

Lab 3.1.1

Write a code that checks if the quadratic equation $x^2 + 3x + 4 = 0$ has one, two or no real solutions.

view answer

Lab 3.1.2

Write a code that checks if the quadratic equation $5x^2 + 3x - 16 = 0$ has one, two or no real solutions.

view answer

3.2 Python Functions

Watch demo video

Example: Given $a, b, c,$ and the quadratic equation $ax^2 + bx + C = 0$. write a `function` to solve such equations. See answer in the code-cell below.

In [ ]:
```
def quadra(a,b,c):
  a,b,c = float(a),float(b),float(c)   # three lines in one
  if b**2-4*a*c < 0:
    return print("No real solutions")
  elif b**2-4*a*c == 0:
    x1 = -b/(2*a)
    return print("the solution is", x1)
  else:
    x1 = (-b-(b**2-4*a*c)**.5)/(2*a)
    x2 = (-b+(b**2-4*a*c)**.5)/(2*a)
    return print("the solutions are: ",x1," and ",x2)
```

Lab 3.2.1

Use the `quadra(a,b,c)` function above to solve $-x^2 - x + 10 = 0$.

view answer

Lab 3.2.2

Use the `quadra(a,b,c)` function to solve $x^2 + x + 1 = 0$.

view answer

3.3 Syntax Errors

Watch demo video

Lab 3.3.1

Identify the error

```
b=2
 c=4
b/c
```

view answer

Lab 3.3.2

Identify the error in the code below:

```
a = 1
b = 2
if a > b
 print("a is bigger than b")
```

```
elif a == b:
 print("equal")
else:
 print("a is smaller than b")
```

view answer

Lab 3.3.3

Identify the error

```
a = 1
b = 2
if a > b:
 print("a is bigger than b")
elif a = b:
 print("equal)
else:
 print("a is smaller than b")
```

view answer

Lab 3.3.4

Identify the error

```
a = 1
b = 2
if a > b:
 print("a is bigger than b")
elif a == b:
 print("equal)
else:
 print("a is smaller than b")
```

view answer

4. Python Loops

There are two type of loops (repeated execution of statements): `for` and `while` loops

4.1 The `for` Loop

Watch demo video 1

Watch demo video 2

Lab 4.1.1

Use the `for` loop to find the average age of the IL Supreme Court Justices given the list of ages: `x=[77, 72,73,62,77,55]`.

view answer

4.2 The `while` Loop

Example

`while` loop for Fibonacci Numbers

Fibonacci Numbers: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, â€¦

The first two terms are 1 and 1. All other terms are obtained by adding the preceding two terms.

Below is a code that lists Fibonacci numbers up to a user given count.

In [ ]:
```
# type in the number of Fibonacci number you want to list and press enter.
n = input("How many Fibonacci numbers? Type a number and press enter. ")

n = int(n)              # convert it to an integer

i = 0                   # set i to zero initially
a = 1                   # first number is 1
b = 1                   # second number is 1

while i < n:            # i starts at zero and ends at n-1
    print (a)
    i = i + 1
    c = a + b          # sum assigned to c
    a = b              # second assigned to first
    b = c              # sum assigned to second
```

```
How many Fibonacci numbers? Type a number and press enter. 5
1
1
2
3
5
```

Lab 4.2.1

List the first 20 Fibonacci numbers.

view answer

Lab 4.2.2

Write a Python code that lists all positive integers starting from 1 till a user given number.

view answer

5. Frequency Distributions and Graphs

5.1 Frequency Distributions

Watch demo video 1

Watch demo video 2

Watch demo video 3

Lab 5.1.1

Write a Python code to find the frequency counts of the following data: `20, 21, 25, 20, 29, 25, 22, 25, 21, 25`

view answer

Lab 5.1.2

20 students were asked if they were registered to vote. Their responses are listed below. Construct a frequency distribution.

```
NO, NO, YES, YES, NO, YES, NO, NO, YES, NO, YES, NO, NO, YES,
                  YES, YES, YES, NO, YES, YES
```

view answer

Lab 5.1.3

Write a Python code to construct a frequency distribution with 5 classes or bins for the following test scores: `40,40,40,40,40,41,41,41,41,41,41,42,42,42,42,43,43,43,43,43,43,44,44,44`

view answer

5.3 Bar Graphs and Pie Charts

Watch demo video 1

Watch demo video 2

Lab 5.3.1

The latest number of coronavirus deaths (July 2021) in Chicago by race is given below. Write a Python code to construct a bar graph.

Race	Deaths
Latinx	1859
Black	2224
White	1233
Other	309

view answer

Lab 5.3.2

The latest number of coronavirus deaths in Chicago by race is given below. Write a Python code to construct a pie chart.

Race	Deaths
Latinx	1859
Black	2224
White	1233
Other	309

view answer

5.4 Histogram, Polygon and Ogive

Watch demo video

Lab 5.4.1

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Write a Python code to construct a histogram with 6 classes(bins).

```
46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                  57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                  70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                  76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                  84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                  92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                  109620, 146868
```

view answer

Lab 5.4.2

Example: A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Write a Python code using `matplotlib` to construct a histogram with 6 classes(bins).

```
46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                  57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                  70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                  76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                  84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                  92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                  109620, 146868
```

view answer

Polygon

Watch demo video

In [ ]:
```
def polygon(x,n):
  import matplotlib.pyplot as plt
  import numpy as np

  freq, bins, edges = plt.hist(x,bins=n,ec="k",range=[min(x),max(x)+1])
  plt.title("Polygon")
  plt.xlabel('data values')
  plt.ylabel('frequency')

  a1=bins[0]-(bins[1]-bins[0])
  a2=bins[-1]+(bins[1]-bins[0])
  bin1=np.insert(bins,[0],a1)
  bin1=np.append(bin1,a2)

  freq,edges,_=plt.hist(x,bins=bin1,ec='k', range=[min(x),max(x)+1])
  midpoints=0.5*(edges[1:]+edges[:-1])
  plt.plot(midpoints,freq)
  plt.xticks(bin1)
  plt.show()
```

Lab 5.4.3

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Use the `polygon` function to construct a polygon with 6 classes(bins).

```
46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                109620, 146868
```

view answer

Ogive

Watch demo video

In [ ]:
```
def ogive(x,n):
  import matplotlib.pyplot as plt
  import numpy as np
```

```
freq, bins, edges = plt.hist(x,bins=n,ec="k",range=[min(x),max(x)+1])
plt.title("Ogive")
plt.xlabel('data values')
plt.ylabel('Cumulative frequency')

a1=bins[0]-(bins[1]-bins[0])
a2=bins[-1]+(bins[1]-bins[0])
bin1=np.insert(bins,[0],a1)
bin1=np.append(bin1,a2)

freq,edges,_=plt.hist(x,bins=bin1,ec='k', range=[min(x),max(x)+1])

ogive1 = edges[:-1]
cumfreq = np.cumsum(freq)
plt.plot(ogive1,cumfreq)
plt.xticks(bin1)
plt.show()
```

Lab 5.4.4

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Use the `ogive` function to construct an ogive with 6 classes(bins).

```
46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                  57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                  70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                  76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                  84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                  92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                  109620, 146868
```

view answer

5.5 Dot Plot

Watch demo video

In [ ]:
```
# dotplot by Patrick FitzGerald
def dotplots(x,x_label="Data Values"):   # function name is dotplots takes a x

 import numpy as np
 import matplotlib.pyplot as plt

 data = np.array(x)   # convert to array
```

```
 values, counts = np.unique(data, return_counts=True)

# dot plot with appropriate figure size and y-axis limits
 fig, ax = plt.subplots(figsize=(6, 2.25))
 for value, count in zip(values, counts):
    ax.plot([value]*count, list(range(count)), 'co', ms=10, linestyle='')
 for spine in ['top', 'right', 'left']:
    ax.spines[spine].set_visible(False)
 ax.yaxis.set_visible(False)
 ax.set_ylim(-1, max(counts))
 ax.set_xticks(range(min(values), max(values)+1))
 ax.tick_params(axis='x', length=0, pad=8, labelsize=12)

 plt.suptitle("Dot Plot")
 plt.ylabel("frequency")
 plt.xlabel("values(categories)")

 return plt.show()
```

Lab 5.5.1

The dataset below is the waiting time, in minutes, of customers at a fast food restaurant. Use the `dotplots` function to construct a dot plot.

```
6, 4, 4, 2, 1, 3, 5, 2, 4, 2, 5, 4, 4, 7, 4, 3, 4, 1, 2,
                  5
```

view answer

5.6 Stem and Leaf Plot

Watch demo video

Lab 5.6.1

Given the test scores listed below, write a Python code to construct and stem and leaf plot.

```
54, 22, 77, 92, 51, 68, 37, 27, 68, 47, 42, 38, 18, 46, 74,
                  77, 81, 78, 16, 35, 77, 90, 82, 36, 50, 39, 19, 62, 70,
88,
                  36, 37, 53, 72, 77
```

view answer

6. Measure of Center and Variation

Watch demo video

6.1 Mean and Median

Lab 6.1.1

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Write a Python code to compute the mean and median.

```
 46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                  57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                  70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                  76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                  84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                  92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                  109620, 146868
```

view answer

6.2 Variance and Standard Deviation

Watch demo video

Lab 6.2.1

The salaries for all the Human Resources department employees of the City of Chicago is listed below. Write a Python code to compute the **population variance and standard deviation**.

```
 75408, 82368, 103968, 82368, 42996, 85992, 67464, 103716,
                  134292, 113484, 94848, 101628, 119712, 61776, 122496,
57720,
                  119712, 119712, 61140, 103716, 67464, 58680, 91020,
```

```
83268,
                    62712, 78828, 82788, 91020, 75408, 59820, 79812, 63348,
                    122496, 72744, 98628, 103968, 103968, 131664, 115788,
76248,
                    119712, 83268, 75408, 75408, 95172, 67824, 75408,
119712,
                    141696, 97392, 83268, 59820, 97668, 82368, 53736, 65676,
                    103716, 94848, 79020, 67824, 86856, 103716, 75408,
119712,
                    91944, 94848, 66336, 99480, 119712
```

view answer

Lab 6.2.2

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Write a Python code to compute the **sample variance and standard deviation**.

```
 46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                    57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                    70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                    76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                    84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                    92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                    109620, 146868
```

view answer

Weighted Mean

Watch demo video

6.4 Grouped Mean and Standard Deviation

Lab 6.4.1

Below is a frequency table of the salaries for the Human Resources department of City of Chicago is listed below. Write a Python code to compute the **sample mean and standard**

deviation.

salaries	freq
5200	9
7200	21
9200	18
11200	16
13500	5

view answer

7. Measures of Position

7.1 Percentiles

Watch demo video

Lab 7.1.1

The salaries for all the Human Resources department of City of Chicago is listed below. Write a Python code to compute the 10th, 25th, 50th, 75th and 90th percentiles.

```
 75408, 82368, 103968, 82368, 42996, 85992, 67464, 103716,
                  134292, 113484, 94848, 101628, 119712, 61776, 122496,
57720,
                  119712, 119712, 61140, 103716, 67464, 58680, 91020,
83268,
                  62712, 78828, 82788, 91020, 75408, 59820, 79812, 63348,
                  122496, 72744, 98628, 103968, 103968, 131664, 115788,
76248,
                  119712, 83268, 75408, 75408, 95172, 67824, 75408,
119712,
                  141696, 97392, 83268, 59820, 97668, 82368, 53736, 65676,
                  103716, 94848, 79020, 67824, 86856, 103716, 75408,
119712,
                  91944, 94848, 66336, 99480, 119712
```

view answer

7.2 SciPy and Percentile Rank

Watch demo video

Lab 7.2.1

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Write a Python code to compute the percentile salary rank of an employee with salary 80000.

```
46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                109620, 146868
```

view answer

7.3 Box Plot

Watch demo video

Lab 7.3.1

The following data is winter temperatures in randomly selected towns and villages in Alaska (simulated data). Write a Python code to draw a box plot.

```
27.3 5.5 14.3 3.2 16.8 17.9, 14.2 14.2 9.6 34.7 8.9 30.1 27.7
                17.7 21.8 47.7 37 24.9
```

view answer

8 Discrete Probability Distributions

8.1 Expected Value and Standard Deviation

Watch demo video

Lab 8.1.1

The random variable X is defined as the number of times a graduating senior changed majors. The probability distribution of X is given below.

|x| 0|1 |2 |3 |4 |5 |6 |7 |8 |p(x) |0.197 |0.212 |0.255 |0.182 |0.09 |0.04 |0.02 |0.003 |0.001

1. What is the mean number of times students changed majors?
2. What is the standard deviation of X?

view answer

8.2 Binomial Probability Distribution

Watch demo video 1

Watch demo video 2

Lab 8.2.1

21% of all Chicagoans live in poverty. If 30 Chicagoans are randomly selected, find the probability that

1. Exactly 1 of them live in poverty
2. At most 4 of them live in poverty
3. At least 4 of them live in poverty

view answer

Binomial Probability Plot

Example

Use the `binomplot` function below to plot a binomial distribution with n = 20 and probability of success is 0.70.

In [ ]:
```
# binomial graph with n and p
def binomplot(n,p):
  from scipy import stats
  import numpy as np
  import matplotlib.pyplot as plt
```

```
x = np.arange(0,n+1)
y = stats.binom.pmf(x,n,p)
plt.bar(x,y,width=0.2)

plt.suptitle("Binomial Plot")
plt.xlabel("x values")
plt.ylabel("probability")

return plt.show()
```

Lab 8.2.2

Use the `binomplot` function to plot a binomial distribution with n = 10 and probability of success is 0.60.

view answer

9. Continuous Probability Distribution

9.1 Uniform Probability Distribution

Lab 9.1.1

Example: The amount of time, in minutes, that a person waits for a train, in a train station, is uniformly distributed between 5 and 15 minutes, inclusive.

- What is the probability that a person waits fewer than 15 minutes?

view answer

9.2 Normal Probability Distribution

Watch demo video

Lab 9.2.1

In a standard normal distribution, where the mean is zero and standard deviation is one, what is the probability of obtaining a value (*z*-value) less than 1.5?

view answer

Watch demo video

Lab 9.2.2

The average college cost per year is `$27191` with the standard deviation of `$7126`. Assume the distribution is normal.

- Find the probability that a randomly selected college will cost less than `$27726` per year.

view answer

Inverse Normal

Watch demo video

Lab 9.2.3

The average college cost per year is $27,191 with standard deviation of $7,126. Assume the distribution is normal.

- Find the 75th percentile for this distribution

view answer

9.3 Central Limit Theorem: mean

Watch demo video

Lab 9.3.1

The lengths of metal screws are normally distributed, with a mean of 5 inches, and standard deviation of 1 inch.

1. If 10 screws are chosen at random, what is the probability that their mean length is less than 6 inches?
2. If 10 screws are chosen at random, what is the probability that their mean length is between 5 and 6 inches?

view answer

Confidence Intervals

10.1 Confidence Intervals for Mean (z distribution)

Watch demo video 1

Watch demo video 2

Lab 10.1.1

Exam scores are normally distributed with a standard deviation of 15 points. The population mean is unknown. A random sample of 50 scores has a mean of 80. Construct a 90% confidence interval for the population mean. Use the `conf_z_mean` function above.

view answer

10.2 Confidence Intervals for Mean (t distribution)

Watch demo video

Lab 10.2.1

A random sample of 25 test scores was selected from an approximately normally distributed population of test scores. The mean and standard deviation are 85 and 7 respectively.

1. Compute the margin of error using the `error_t_mean` fucntion above for constructing a 90% confidence interval.
2. Construct a 90% confidence interval using the `conf_t_mean` function above.

view answer

10.3 Confidence Intervals for Proportions (z distribution)

Watch demo video

Lab 10.3.1

In a random sample of 120 adults, 51 are fully vaccinated for Covid19. Construct a 90% confidence interval using the `conf_prop` function above.

view answer

Lab 10.3.2

In a random sample of 120 adults, 50% are fully vaccinated for Covid19. Construct a 95% confidence interval using the `conf_prop` function above.

view answer

Lab 10.3.3

Town officials want to determine the current percentage of residents vaccinated for Covid19. How many residents should be surveyed in order to be 90% confident that the estimated proportion is within five percentage points of the true population proportion who are vaccinated for Covid19? Previously the proportion was 0.55. Use the `samp_pro` function below.

view answer

11. Hypothesis Testing

11.1 Hypothesis Testing: Mean, z-test

Watch demo video

Lab 11.1.1

A government official claims that the average recovery time for Covid19 patients with mild symptoms is less than 2 weeks. A random sample of 25 patients with mild symptoms has a mean recovery time of 2.4 weeks with standard deviation of 0.9 weeks. Assume recovery time is normally distributed with standard deviation of 0.5 (σ) weeks. Test the government's official claim at Î± of 0.05. Use the `z_test_mean` function above.

view answer

11.2 Hypothesis Testing: Mean, t-test

Watch demo video

Lab 11.2.1

A government official claims that the average recovery time for Covid19 patients with mild symptoms is 1.5 weeks. A random sample of 25 patients with mild symptoms. Their recovery time is given below. Assume recovery time is normally distributed with standard deviation of 0.5 (Ïf) weeks. Test the governmentâ€™s official claim at Î± of 0.05. Use the `z_test_mean` function above.

```
2.5, 3.0, 1.5, 2.4, 2.6, 2.1, 1.8, 1.5, 1.7, 1.9, 3.2, 2.3,
                  2.0, 1.8, 1.2, 1.5, 2.3, 2.5, 1.8, 2.2, 3.3, 2.3, 1.6,
2.5,
                  2.2
```

view answer

12.3 Hypothesis Testing: Proportions, z-test

Watch demo video

Lab 12.3.1

The President of USA believes that more than 50% of the US population are fully vaccinated for Covid19. In a survey of 1000 residents, 510 were fully vaccinated. Test the president's claim at α level of 0.05. Use the `z_test_prop` function.

view answer

12. Linear Regression and Correlation

12.1 Scatter Plot and Correlation

Demo Vidoe Link Scatter Plot

Watch demo video 2 Correlation Coefficient

Lab 12.1.1

The number of years (age) x and prices y, for 10 cars is listed below. Compute the correlation coefficient and display the scatter plot.

x	1	0	3	5	14	7	4	16	9	10
y	17000	20000	15000	10000	5000	9000	10000	2000	8000	9000

view answer

12.3 Linear Regression

Watch demo video

Lab 12.3.1

The number of years (age) x and prices y, for 10 cars is listed below. Compute the correlation coefficient (r-value), estimated slope and y-intercept of the regression line and the p-value

for testing the significance of correlation.

x	1	0	3	5	14	7	4	16	9	10
y	17000	20000	15000	10000	5000	9000	10000	2000	8000	9000

view answer

13. Analysis of Categorical Data

13.1 Goodness of Fit

Watch demo video

Lab 13.1.1

The table below shows the distribution of the 2020 US presidential electoral votes. An official claims that the distribution is proportional with the popular vote, with 51.3% for Biden and 46.9% for Trump. Test the officials claim at $\alpha = 0.10$.

Candidate	Electoral-Votes	Popular vote
Biden	306	51.3%
Trump	232	46.9%

view answer

13.2 Test of Independence

Watch demo video

Lab 13.2.1

The contingency table below shows the prison population of Illinois in June 2021. Test the hypothesis that race (Black, White and Hispanic) and crime-class are independent at 5% level of significance.

Class|Black|Hispanic|White ---|---|---|---| 1|1734|475|1281 2|2071|373|1938 3|678|103|935 4|534|92|560| X|5644|1661|2613 Murder|4094|927|1277

Use the `chi2_test_of_indp` function above.

view answer

14. Advanced Statistics Topics

14.1 Descriptive Statistics with Pandas

Watch demo video

Pandas Library

Lab 14.1.1

Write a Python code to convert the dataset below to pandas data frame with variable name `gender`.

```
x = ['male','female','female','male','male']
```

view answer

Frequency and Relative Frequency with Pandas

Lab 14.1.2

Write a Python code using pandas to find the frequency counts of the following data: `20, 21, 25, 20, 29, 25, 22, 25, 21, 25`

view answer

Lab 14.1.3

Write a Python code using pandas to find the *relative* frequency counts of the following data: `20, 21, 25, 20, 29, 25, 22, 25, 21, 25`

view answer

Lab 14.1.4

20 students were asked if they were registered to vote. Their responses are listed below. Construct a frequency distribution using pandas.

```
NO, NO, YES, YES, NO, YES, NO, NO, YES, NO, YES, NO, NO, YES,
                  YES, YES, YES, NO, YES, YES
```

view answer

Summary Statistics

Lab 14.1.5

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Write a Python code using `pandas` to compute summary statistics: mean, standard deviation, minimum, 1st quartile, 2nd quartile, 3rd quartile and maximum salary.

```
 46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                  57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                  70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                  76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                  84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                  92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                  109620, 146868
```

view answer

Skewness and Kurtosis

Lab 14.1.6

A random sample of 54 annual salaries of Aviation employees of the City of Chicago is listed below. Write a Python code with `pandas` to compute the skewness and kurtosis values.

```
 46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408,
                  57408, 57408, 60108, 60108, 62712, 63552, 66852, 66852,
69732,
                  70032, 70044, 70272, 72024, 73380, 73380, 73380, 75408,
75468,
                  76248, 80484, 80484, 80484, 82476, 82788, 82836, 83676,
84324,
                  84324, 84324, 85848, 87564, 88272, 88272, 90828, 92004,
92304,
                  92520, 96060, 96096, 100680, 100716, 100716, 103764,
105420,
                  109620, 146868
```

view answer

14.2 Uploading Dataset

Watch demo video Read CSV from URL

Watch demo video Upload Excel from Computer

Lab 14.2.1

Write a Python code to read a dataset in csv format from a website (URL) into Google Colab. Use the URL below. Convert the dataset to pandas dateframe and print basic information about the data set using `x_df.info()` method.

https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/master/data.csv

view answer

14.3 Descriptive Statistics (multiple columns)

Watch demo video

Lab 14.3.1

Write a Python code to read the dataset with the url below and find the mean, median, standard deviation, minimum, maximum, 25th and 75th percentiles (descriptive statistics) of all columns in the dataset.

https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/master/data.csv

view answer

Lab 14.3.2

Write a Python code to read the dataset with url below and find the mean, std, and size of the `actual_weight` grouped by `country_of_origin`.

'https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/master/data.csv'

view answer

Lab 14.3.3

Write a Python code to find the mean, std, size of the `actual_weight` grouped by `country_of_origin` and `route`.

https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/master/data.csv

view answer

Lab 14.3.4

Write a Python code to display a boxplot of the `actual_weigth` grouped by `country_of_origin`.

https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/master/data.csv

view answer

14.4 Visualization

Lab 14.4.1

The dataset below (URL) contains students test1_score, test2_score,test3_score, gender, level, section etc.

Write a Python code to display a frequency bar graph for `section`.

https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv

view answer

Lab 14.4.2

The dataset below (URL) contains students test1_score, test2_score,test3_score, gender, level, section etc.

Write a Python code to display a frequency pie chart for `section`.

https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv

view answer

14.5 Regression

Multiple Linear Regression

Watch demo video

Lab 14.5.1

The dataset in the website (URL below) includes 20 columns about home sale, including price, number of bedrooms and bathrooms, areas etc. The data includes 21,613 entries.

1. Write a Python code to find the least-squares prediction equation, $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ for these data where, y is `price`, x_1 is `bedrooms`, x_2 `sqft_basement` and x_3 is `sqft_living`.
2. Use the overall F test to determine whether the model contributes significant information to the prediction of y with 1% level of significance.
3. What is the value of R-squared. That is, the coefficient of determination.
4. Does the `sqft_living`, x_3, contribute significant information for the prediction of the price, y, given that x_1 and x_2 are already in the model? (use $\alpha = 0.05$)

https://raw.githubusercontent.com/Shreyas3108/house-price-prediction/master/kc_house_data.csv

view answer

Logistic Regression

Lab 14.5.2

The data set below (URL) contains the occurrence of coronary heart disease, chd, tobacco, obesity, age, alcohol, etc. For chd, 0 means patient has no chd and 1 means patient has chd.

1. Write a Python code to read the data set and display the first five rows.
2. Write a Python code to fit a logistic regression model where the dependent variable is `chd` and the predictor variables are `ldl`, `sbs` and `age`. Use 5% level of significance.

https://raw.githubusercontent.com/simeonaman/demo/master/heart.csv

view answer

14.6 ANOVA

Watch demo video

One-way ANOVA

Lab 14.6.1

The data set below (URL) contains the students test1_score, gender, level, section etc.

1. Write a Python code to read the data set and display the first and last five rows.
2. Write a Python code to find `test3_score` means, std and size for each level
3. Write a Python code to perform ANOVA to test if there is sufficient evidence to indicate a difference in the averages of `test1_scores` among the three `levels` (low, medium, high). Use 10% level of significance.

https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv

view answer

Two-way ANOVA (Block Design)

Lab 14.6.2

The data set below (URL) contains students test1_score, gender, level, section etc.

1. Write a Python code to read the dataset and display the first five rows.
2. Write a program to find `test3_score` means grouped by `level` and `gender`.
3. Write a Python code to perform ANOVA block design to test if there is sufficient evidence to indicate a difference in the averages of `test3_score` among the three levels (low, medium, high) using gender as blocks. Use 10% level of significance.

https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv

view answer

Two-Factor ANOVA (Interactions)

Lab 14.6.3

The dataset below (URL) contains students `test1_score`, `gender`, `level`, `section` etc.

1. Write a Python code to read the dataset and display the first 5 rows.
2. Write a Python code to display the means grouped by `level` and `interaction` for `test3_score`.
3. Write a Python code to perform a two-factor ANOVA with `level` and `section`. Do the data provide sufficient evidence to indicate an interaction between `level` and `section`? Use 10% level of significance.

https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv

view answer

Lab Answers

Lab 1.1.1

In [ ]:
```
print("Hello, World")
```

```
Hello, World
```

Lab 1.1.2

In [ ]:
```
print('Welcome to Python Programming Language.')
```

```
Welcome to Python Programming Language.
```

Lab 1.2.1

In [ ]:
```
10+10
```

Out[ ]:
```
20
```

Lab 1.2.2

In [ ]:
```
100-88
```

Out[ ]:
```
12
```

Lab 1.2.3

In [ ]:
```
1/2+7.1-10*3
```

Out[ ]:
```
-22.4
```

Lab 1.2.4

In [ ]:
```
-5 + 7 - (100+.05)/10
```

Out[ ]:
```
-8.004999999999999
```

Lab 1.2.5

In [ ]:
```
1/671000000
```

Out[ ]:
```
1.4903129657228019e-09
```

Lab 1.2.6

In [ ]:
```
6.25*100*16000
```

Out[ ]:
```
10000000.0
```

Lab 1.2.7

In [ ]:
```
28/4*2
```

Out[ ]:
```
14.0
```

Lab 1.2.8

In [ ]:
```
7**2+1/2**(1/2)
```

Out[ ]:
```
49.707106781186546
```

Lab 1.2.9

In [ ]:
```
7**2-8**(1/2)
```

Out[ ]:
```
46.17157287525381
```

Lab 1.2.10

In [ ]:
```
(7**2-8**(1/2))/25
```

Out[ ]:
```
1.8468629150101523
```

Lab 1.3.1

In [ ]:
```
x1 = 6
y1 = 9
x2 = 5
y2 = 1
```

```
m = (y2-y1)/(x2-x1)

x = (x1+x2)/2
y = (y1+y2)/2

print('slope is',m)
print('mid point is (',x,',',y,')')
```

```
slope is 8.0
mid point is ( 5.5 , 5.0 )
```

Lab 1.3.2

In [ ]:
```
a = 2
b = -3
c = -4

x1 = -b-(b**2-4*a*c)**(1/2)
x1 = x1/(2*a)

x2 = -b+(b**2-4*a*c)**(1/2)
x2 = x2/(2*a)

print("one solution is ", x1)
print("another solution is ", x2)
```

```
one solution is  -0.8507810593582121
another solution is  2.350781059358212
```

Lab 1.3.3

In [ ]:
```
round((1+2)/18,1)
```

Out[ ]:
```
0.2
```

OR

In [ ]:
```
a = (1+2)/18
round(a,1)
```

Out[ ]:
```
0.2
```

Lab 1.4.1

In [ ]:
```
length = 5
width = 8
```

```
area = length*width  # the area of a rectangle is length times width
print(area)         # display the result
```

```
40
```

Lab 1.5.1

In [ ]:
```
abs(7*2-89**(1/2))
```

Out[ ]:
```
4.566018867943397
```

OR

In [ ]:
```
a = 7*2-89**(1/2)
abs(a)
```

Out[ ]:
```
4.566018867943397
```

Lab 1.5.2

In [ ]:
```
(abs(7**2-8**(1/2)))/25
```

Out[ ]:
```
1.8468629150101523
```

OR

In [ ]:
```
a = 7**2-8**(1/2)
b = abs(a)
b/25
```

Out[ ]:
```
1.8468629150101523
```

Lab 1.5.3

In [ ]:
```
import numpy as np

sin = np.sin(np.pi/3)

Area = 10*12*sin

print(Area)
```

```
103.92304845413263
```

Lab 1.5.4

In [ ]:
```
import numpy as np

np.sqrt(6**2-4*2*(-5))
```

Out[ ]:
```
8.717797887081348
```

OR

In [ ]:
```
import numpy as np

a = 6**2-4*2*(-5)
np.sqrt(a)
```

Out[ ]:
```
8.717797887081348
```

Lab 1.6.1

In [ ]:
```
w=input("Enter weight in pounds:")  #ask the user for a number
h1 = input("Enter height feet only:")
h2 = input("Enter inches:")
w,h1,h2 = float(w),float(h1),float(h2) # converst the number to float
bmi = 703*(w/(h1*12+h2)**2)
print("The BMI is", bmi)
```

```
Enter weight in pounds:180
Enter height feet only:5
Enter inches:9
The BMI is 26.578449905482042
```

Lab 2.1.1

In [ ]:
```
x = [1,2,3,4,5,6,7,8,9]
x
```

Out[ ]:
```
[1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Lab 2.1.2

In [ ]:
```
x = [1,2,3,4,5,6,7,8,9]
x[0]
```

Out[ ]:
```
1
```

Lab 2.1.3

In [ ]:
```
x = [1,2,3,4,5,6,7,8,9]
x[-1]
```

Out[ ]:
```
9
```

Lab 2.1.4

In [ ]:
```
x = [1,2,3,4,5,6,7,8,9]
x[1:4]
```

Out[ ]:
```
[2, 3, 4]
```

Lab 2.2.1

In [ ]:
```
x = [72,82,66,70,66,60,53,56,49]

x.sort()
print(x)
```

```
[49, 53, 56, 60, 66, 66, 70, 72, 82]
```

Lab 2.2.2

In [ ]:
```
x = [72,82,66,70,66,60,53,56,49]

max(x)
```

Out[ ]:
```
82
```

Lab 2.2.3

In [ ]:
```
x = [72,82,66,70,66,60,53,56,49]

x.sort()
x.reverse()
```

In [ ]:
```
print(x)
```

```
[82, 72, 70, 66, 66, 60, 56, 53, 49]
```

Lab 3.1.1

In [ ]:
```
a = 1   # coefficient of x^2
b = 3   # coefficient of x
c = 4  # constant term

d = b**2 - 4*a*c

if d < 0:
  print("No real solution")
elif d == 0:
  print("One real solution")
else:
  print("Two real solutions")
```

```
No real solution
```

Lab 3.1.2

In [ ]:
```
a = 5   # coefficient of x^2
b = 3   # coefficient of x
c = -16  # constant term

d = b**2 - 4*a*c

if d < 0:
  print("No real solution")
elif d == 0:
  print("One real solution")
else:
  print("Two real solutions")
```

```
Two real solutions
```

Lab 3.2.1

In [ ]:
```
def quadra(a,b,c):
  a,b,c = float(a),float(b),float(c)
  if b**2-4*a*c < 0:
    return print("No real solutions")
  elif b**2-4*a*c == 0:
    x1 = -b/(2*a)
    return print("the solution is", x1)
  else:
    x1 = (-b-(b**2-4*a*c)**.5)/(2*a)
    x2 = (-b+(b**2-4*a*c)**.5)/(2*a)
    return print("the solutions are: ",x1," and ",x2)
```

In [ ]:
```
quadra(-1,-1,10)  # in this case a=-1 b=-1 c=10
```

```
the solutions are:  2.7015621187164243  and  -3.7015621187164243
```

Lab 3.2.2

In [ ]:
```
def quadra(a,b,c):
  a,b,c = float(a),float(b),float(c)
  if b**2-4*a*c < 0:
    return print("No real solutions")
  elif b**2-4*a*c == 0:
    x1 = -b/(2*a)
    return print("the solution is", x1)
  else:
    x1 = (-b-(b**2-4*a*c)**.5)/(2*a)
    x2 = (-b+(b**2-4*a*c)**.5)/(2*a)
    return print("the solutions are: ",x1," and ",x2)
```

In [ ]:
```
quadra(1,1,1)   # heare a=1 b=1 c=1
```

```
No real solutions
```

Lab 3.3.1

In [ ]:
```
b=2
 c=4
b/c
```

```
  File "<ipython-input-86-158994d5983a>", line 2
    c=4
    ^
IndentationError: unexpected indent
```

Lab 3.3.2 `if a > b:`

In [ ]:
```
a = 1
b = 2
if a > b
 print("a is bigger than b")
elif a = b:
 print("equal)
else:
 print("a is smaller than b")
```

```
  File "<ipython-input-87-871e93d2a39b>", line 3
    if a > b
           ^
SyntaxError: invalid syntax
```

Lab 3.3.3

In [ ]:
```
a = 1
b = 2
```

```
if a > b:
 print("a is bigger than b")
elif a = b:
 print("equal)
else:
 print("a is smaller than b"):
```

```
  File "<ipython-input-88-128f45613308>", line 5
    elif a = b:
           ^
SyntaxError: invalid syntax
```

Lab 3.3.4

In [ ]:
```
a = 1
b = 2
if a > b:
 print("a is bigger than b")
elif a == b:
 print("equal)
else:
 print("a is smaller than b")
```

```
  File "<ipython-input-90-9aa3a9224cf2>", line 6
    print("equal)
                 ^
SyntaxError: EOL while scanning string literal
```

Lab 4.1.1

In [ ]:
```
age = [77, 72,73,62,77,55]

sum = 0

for i in age:
  sum = sum + i

aver = sum/len(age)

print("The average age of IL Supreme Court Justices is ", aver)
```

```
The average age of IL Supreme Court Justices is  69.33333333333333
```

Lab 4.2.1

In [ ]:
```
n = 20
```

```
i = 0
a = 1
b = 1

while i < n:
    print (a)
    i = i + 1
    c = a + b
    a = b
    b = c
```

```
1
1
2
3
5
8
13
21
34
55
89
144
233
377
610
987
1597
2584
4181
6765
```

Lab 4.2.2

In [ ]:
```
n = input("Type in a maximum positive integer and press enter" )
n = int(n)

i = 1

while i < n+1:
  print(i)
  i = i+1
```

```
Type in a maximum positive integer and press enter5
1
2
3
4
5
```

Lab 4.2.3

In [ ]:
```
n = input("Tpye in a positive integer and press enter ")
n = int(n)

if n == 0 or n == 1:
 f = 1
elif n < 0:
 print("Error: non-negative integers only")
else:
  i = 1
  f = 1
  while i < n+1:
    f = f*i
    i = i + 1
  print("n factorial is",f)
```

```
Tpye in a positive integer and press enter 10
n factorial is 3628800
```

Lab 5.1.1

In [ ]:
```
import numpy as np

x = [20, 21, 25, 20, 29, 25, 22, 25, 21, 25]

np.unique(x,return_counts=True)
```

Out[ ]:
```
(array([20, 21, 22, 25, 29]), array([2, 2, 1, 4, 1]))
```

Lab 5.1.2

In [ ]:
```
import numpy as np

x = ['NO', 'NO', 'YES', 'YES', 'NO', 'YES', 'NO', 'NO', 'YES', 'NO', 'YES',
     'NO', 'NO', 'YES', 'YES', 'YES', 'YES', 'NO', 'YES', 'YES']
np.unique(x,return_counts=True)
```

Out[ ]:
```
(array(['NO', 'YES'], dtype='<U3'), array([ 9, 11]))
```

Lab 5.1.3

In [ ]:
```
import numpy as np

x = [40,40,40,40,40,41,41,41,41,41,41,42,42,42,42,43,43,43,43,43,43,44,44,44,
     44,45]

freq,bin_edges = np.histogram(x,5,density=False)
```

```
print(bin_edges)
print(freq)
```

```
[40. 41. 42. 43. 44. 45.]
[5 6 4 6 5]
```

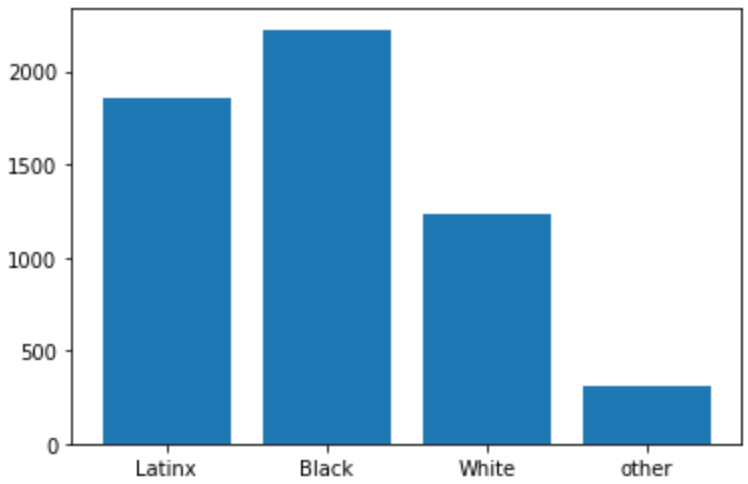
40 to 41 has 5 scores 41 to 42 has 6 scores 42 to 43 has 4 scores 43 to 44 has 6 scores 44 to 45 has 5 scores

scores	freq
40-41	5
41-42	6
42-43	4
43-44	6
44-45	5

Lab 5.3.1

In [ ]:
```
import matplotlib.pyplot as plt


race = ['Latinx','Black','White','other']
death = [1859,2224,1233,309]

plt.bar(race,death)
```

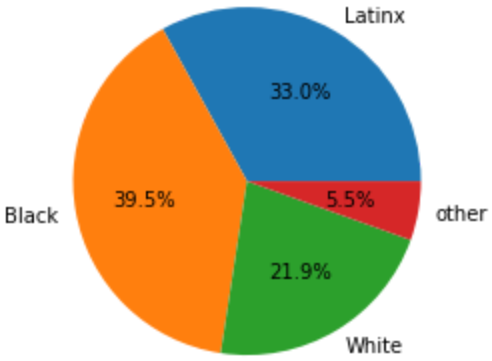
Out[ ]:
```
<BarContainer object of 4 artists>
```



5.3.2

In [ ]:
```
import matplotlib.pyplot as plt

race = ['Latinx','Black','White','other']
death = [1859,2224,1233,309]

plt.pie(death,labels=race,autopct='%1.1f%%')
```

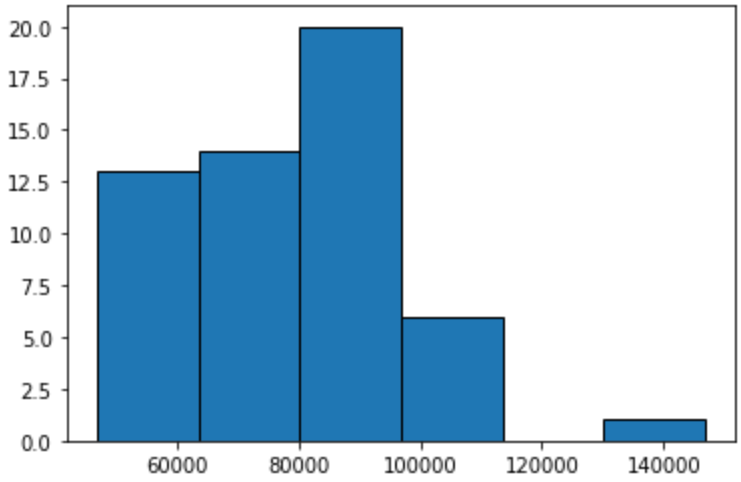
Out[ ]:
```
([<matplotlib.patches.Wedge at 0x7fdd012ef410>,
  <matplotlib.patches.Wedge at 0x7fdd012efb50>,
  <matplotlib.patches.Wedge at 0x7fdd012f9410>,
  <matplotlib.patches.Wedge at 0x7fdd012f9d10>],
 [Text(0.5584906875918265, 0.9476751299222794, 'Latinx'),
  Text(-1.0828050332538752, -0.19373502512471624, 'Black'),
  Text(0.5627194086838145, -0.9451702847056388, 'White'),
  Text(1.0836598543140907, -0.18889499767850831, 'other')],
 [Text(0.3046312841409962, 0.5169137072303341, '33.0%'),
  Text(-0.5906209272293864, -0.10567365006802702, '39.5%'),
  Text(0.3069378592820806, -0.5155474280212574, '21.9%'),
  Text(0.5910871932622312, -0.10303363509736815, '5.5%')])
```



Lab 5.4.1

In [ ]:
```
import matplotlib.pyplot as plt


x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
     72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
     82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
     88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
     103764, 105420, 109620, 146868]

plt.hist(x,bins=6,ec='k')
```

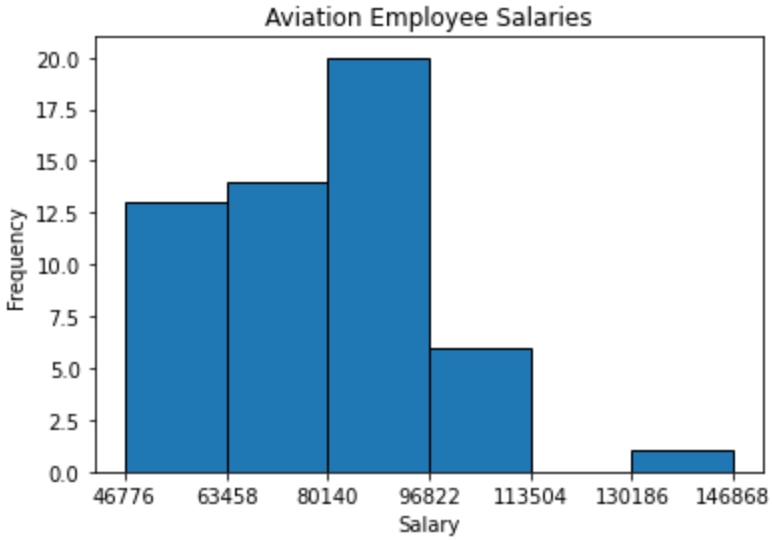
Out[ ]:
```
(array([13., 14., 20.,  6.,  0.,  1.]),
 array([ 46776.,  63458.,  80140.,  96822., 113504., 130186., 146868.]),
 <a list of 6 Patch objects>)
```



Lab 5.4.2

In [ ]:
```
import matplotlib.pyplot as plt

x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
     72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
     82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
     88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
     103764, 105420, 109620, 146868]

freq,bins,_ = plt.hist(x,bins=6,ec='k')
plt.xticks(bins)
plt.title("Aviation Employee Salaries")
plt.xlabel("Salary")
plt.ylabel("Frequency")
```

Out[ ]:
```
Text(0, 0.5, 'Frequency')
```

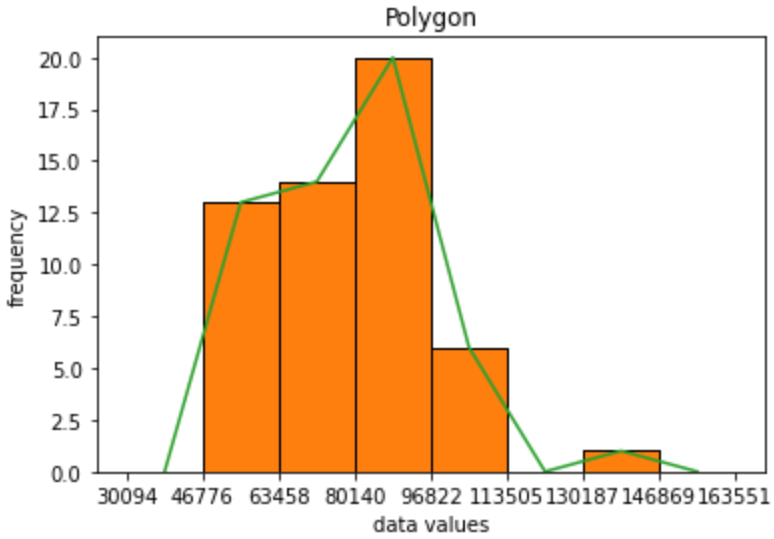


Lab 5.4.3

Copy, paste and run the `polygon` function below. Then write a Python code to call the function with the given data.

In [ ]:
```
def polygon(x,n):
  import matplotlib.pyplot as plt
  import numpy as np

  freq, bins, edges = plt.hist(x,bins=n,ec="k",range=[min(x),max(x)+1])
  plt.title("Polygon")
  plt.xlabel('data values')
  plt.ylabel('frequency')

  a1=bins[0]-(bins[1]-bins[0])
  a2=bins[-1]+(bins[1]-bins[0])
  bin1=np.insert(bins,[0],a1)
  bin1=np.append(bin1,a2)

  freq,edges,_=plt.hist(x,bins=bin1,ec='k', range=[min(x),max(x)+1])
  midpoints=0.5*(edges[1:]+edges[:-1])
  plt.plot(midpoints,freq)
  plt.xticks(bin1)
  plt.show()
```

In [ ]:
```
x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
     72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
     82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
     88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
     103764, 105420, 109620, 146868]

polygon(x,6)
```

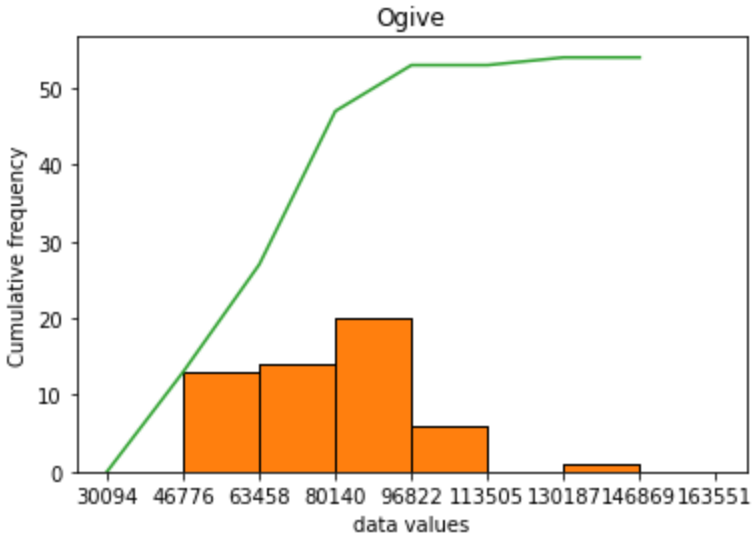


Lab 5.4.4

Copy the 'ogive' function, paste and run it below. Then use it to make an ogive.

In [ ]:
```
def ogive(x,n):
  import matplotlib.pyplot as plt
  import numpy as np

  freq, bins, edges = plt.hist(x,bins=n,ec="k",range=[min(x),max(x)+1])
  plt.title("Ogive")
  plt.xlabel('data values')
  plt.ylabel('Cumulative frequency')

  a1=bins[0]-(bins[1]-bins[0])
  a2=bins[-1]+(bins[1]-bins[0])
  bin1=np.insert(bins,[0],a1)
  bin1=np.append(bin1,a2)

  freq,edges,_=plt.hist(x,bins=bin1,ec='k', range=[min(x),max(x)+1])

  ogive1 = edges[:-1]
  cumfreq = np.cumsum(freq)
  plt.plot(ogive1,cumfreq)
  plt.xticks(bin1)
  plt.show()
```

In [ ]:
```
x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
     72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
     82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
     88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
     103764, 105420, 109620, 146868]

ogive(x,6)
```



Lab 5.5.1

Copy the `dotplots` function and paste it below. Then use it to draw a dot plot.

In [ ]:
```
# dotplot by Patrick FitzGerald
def dotplots(x,x_label="Data Values"):   # function name is dotplots takes a x

 import numpy as np
 import matplotlib.pyplot as plt

 data = np.array(x)   # convert to array
 values, counts = np.unique(data, return_counts=True)

# dot plot with appropriate figure size and y-axis limits
 fig, ax = plt.subplots(figsize=(6, 2.25))
 for value, count in zip(values, counts):
    ax.plot([value]*count, list(range(count)), 'co', ms=10, linestyle='')
 for spine in ['top', 'right', 'left']:
    ax.spines[spine].set_visible(False)
 ax.yaxis.set_visible(False)
 ax.set_ylim(-1, max(counts))
 ax.set_xticks(range(min(values), max(values)+1))
 ax.tick_params(axis='x', length=0, pad=8, labelsize=12)

 plt.suptitle("Dot Plot")
 plt.ylabel("frequency")
 plt.xlabel(x_label)

 return plt.show()
```

In [ ]:
```
x = [6, 4, 4, 2, 1, 3, 5, 2, 4, 2, 5, 4, 4, 7, 4, 3, 4, 1, 2, 5]
dotplots(x, "Waiting Time")
```

Lab 5.6.1

First install stemgraphic

In [ ]:
```
pip install stemgraphic
```

In [ ]:
```
import stemgraphic

x = [54, 22, 77, 92, 51, 68, 37, 27, 68, 47, 42, 38, 18, 46, 74, 77, 81, 78, 16,
     35, 77, 90, 82, 36, 50, 39, 19, 62, 70, 88, 36, 37, 53, 72, 77]
stemgraphic.stem_graphic(x,scale=10,aggregation=False)
```

Out[ ]:
```
(<Figure size 540x216 with 1 Axes>,
 <matplotlib.axes._axes.Axes at 0x7fefe45b7410>)
```

Lab 6.1.1

In [ ]:
```
import numpy as np

x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
     72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
     82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
     88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
     103764, 105420, 109620, 146868]


print("mean and median")
print(np.mean(x))
print(np.median(x))
```

```
mean and median
78460.0
78366.0
```

Interpretation: The mean and median are 78460.0 and 78366.0 respectively.

Lab 6.2.1

In [ ]:
```
import numpy as np

x=[75408, 82368, 103968, 82368, 42996, 85992, 67464, 103716, 134292, 113484,
   94848, 101628, 119712, 61776, 122496, 57720, 119712, 119712, 61140, 103716,
```

```
   67464, 58680, 91020, 83268, 62712, 78828, 82788, 91020, 75408, 59820, 79812,
   63348, 122496, 72744, 98628, 103968, 103968, 131664, 115788, 76248, 119712,
   83268, 75408, 75408, 95172, 67824, 75408, 119712, 141696, 97392, 83268,
   59820, 97668, 82368, 53736, 65676, 103716, 94848, 79020, 67824, 86856,
   103716, 75408, 119712, 91944, 94848, 66336, 99480, 119712]

print('the population variance and standared devaition are')
print(np.var(x,ddof=0))
print(np.std(x,ddof=0))
```

```
the population variance and standared devaition are
490101843.4177693
22138.24390998006
```

Lab 6.2.2

In [ ]:
```
import numpy as np

x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
     72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
     82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
     88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
     103764, 105420, 109620, 146868]

print('The sample variance and standard deviation')
print(np.var(x,ddof=1))
print(np.std(x,ddof=1))
```

```
The sample variance and standard deviation
333192547.0188679
18253.562584297564
```

Interpretation: The sample variance and standard devation are 333192547.0188679 and 18253.562584297564 respectively.

Lab 6.4.3

In [ ]:
```
import numpy as np

x = [5200, 7200, 9200, 11200, 13500]
f = [9,21,18,16,5]

np.average(x,weights=f)
```

Out[ ]:
```
8844.927536231884
```

Interpretaton: The sample mean is 8844.927536231884

In [ ]:
```
import numpy as np

x = [5200, 7200, 9200, 11200, 13500]
f = [9,21,18,16,5]

np.sqrt(np.cov(x,fweights=f,ddof=1))
```

Out[ ]:
```
2350.6905263250064
```

Interpretation: The sample standard devition 2350.6905263250064

Lab 7.1.1

In [ ]:
```
import numpy as np

x = [75408, 82368, 103968, 82368, 42996, 85992, 67464, 103716, 134292, 113484,
     94848, 101628, 119712, 61776, 122496, 57720, 119712, 119712, 61140, 103716,
     67464, 58680, 91020, 83268, 62712, 78828, 82788, 91020, 75408, 59820,
     79812, 63348, 122496, 72744, 98628, 103968, 103968, 131664, 115788, 76248,
     119712, 83268, 75408, 75408, 95172, 67824, 75408, 119712, 141696, 97392,
     83268, 59820, 97668, 82368, 53736, 65676, 103716, 94848, 79020, 67824,
     86856, 103716, 75408, 119712, 91944, 94848, 66336, 99480, 119712]

np.quantile(x,[.10,.25,.50,.75,.90])
```

Out[ ]:
```
array([ 61648.8,  75408. ,  85992. , 103716. , 119712. ])
```

Interpretation: The 10th, 25th, 50th, 75th and 90th percentile are 61648.8, 75408.0, 85992.0, 103716.0, and 119712.0 respectively.

Lab 7.2.1

In [ ]:
```
import numpy as np
from scipy import stats

x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
       60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
       72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
       82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
       88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
       103764, 105420, 109620, 146868]

x_array = np.array(x)   # convert the list to a numpy array

stats.percentileofscore(x_array,80000,kind='rank')
```

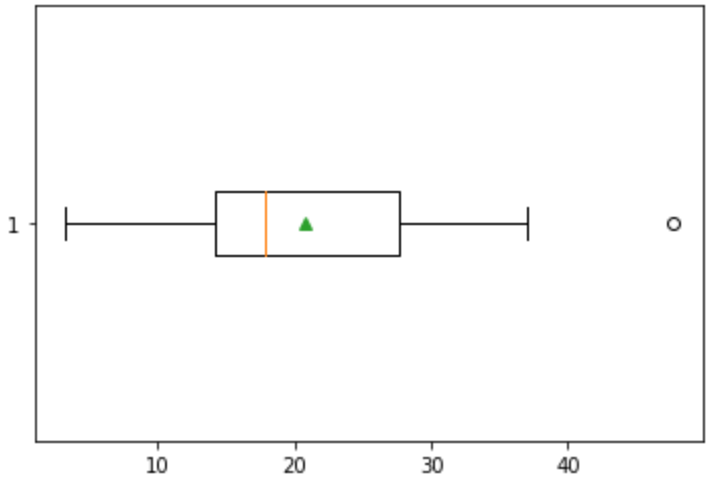
Out[ ]:
```
50.0
```

Interpretation: The percentile salary rank of an emplyee with salary 80000 is 50.

Lab 7.3.1

In [ ]:
```
import matplotlib.pyplot as plt
x = [27.3, 5.5, 14.3, 3.2, 16.8, 17.9, 14.2, 14.2, 9.6, 34.7, 8.9, 30.1,
       27.7, 17.7, 21.8, 47.7, 37, 24.9]
plt.boxplot(x,vert=False,showmeans=True)
```

Out[ ]:
```
{'boxes': [<matplotlib.lines.Line2D at 0x7fdd00f4be90>],
 'caps': [<matplotlib.lines.Line2D at 0x7fdd00ed1f10>,
  <matplotlib.lines.Line2D at 0x7fdd00ed7490>],
 'fliers': [<matplotlib.lines.Line2D at 0x7fdd00ede490>],
 'means': [<matplotlib.lines.Line2D at 0x7fdd00ed7f50>],
 'medians': [<matplotlib.lines.Line2D at 0x7fdd00ed7a10>],
 'whiskers': [<matplotlib.lines.Line2D at 0x7fdd00ed1490>,
  <matplotlib.lines.Line2D at 0x7fdd00ed19d0>]}
```



Lab 8.1.1

In [ ]:
```
import numpy as np

x = [0, 1, 2, 3, 4, 5, 6, 7, 8]
p = [0.197, 0.212, 0.255,       0.182, 0.09, 0.04, 0.02, 0.003, 0.001]

average = np.average(x, weights=p)
std = np.sqrt(np.cov(x,aweights=p, ddof=0))

print('mean is',average)
print('standard deviation is',std)
```

```
mean is 1.977
standard deviation is 1.5272429407268509
```

Lab 8.2.1

In [ ]:
```
from scipy import stats
stats.binom.pmf(1,30,.21)
```

Out[ ]:
```
0.006769026228180294
```

In [ ]:
```
from scipy import stats
stats.binom.cdf(4,30,.21)
```

Out[ ]:
```
0.2145879371696642
```

In [ ]:
```
from scipy import stats
1-stats.binom.cdf(3,30,.21)
```

Out[ ]:
```
0.9015599620024068
```

Interpretation:

$P(x = 1)$ = 0.006769026228180294

$P(x \leq 4)$ = 0.2145879371696642

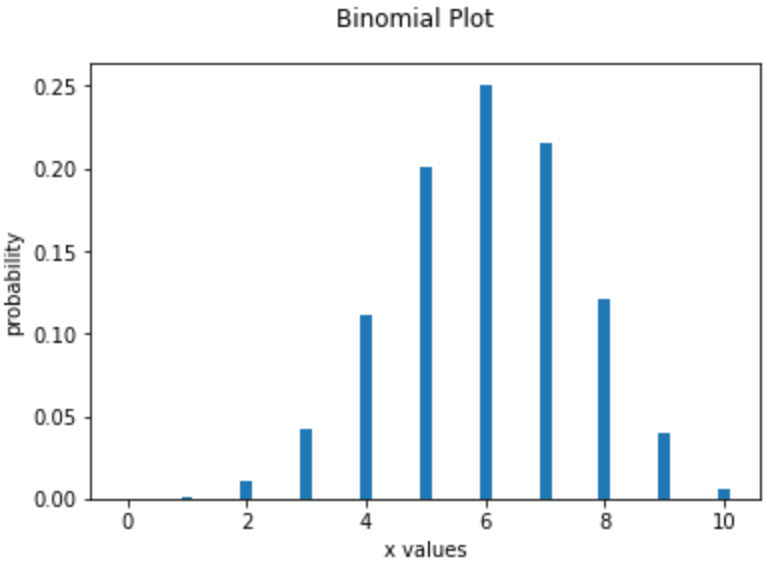
$P(x \geq 4)$ = 0.7854120628303358

Mean = 168.0 and std = 11.520416659131735

Lab 8.2.2

In [ ]:
```
# binomial graph with n and p
def binomplot(n,p):
  from scipy import stats
  import numpy as np
  import matplotlib.pyplot as plt

  x = np.arange(0,n+1)
  y = stats.binom.pmf(x,n,p)
  plt.bar(x,y,width=0.2)

  plt.suptitle("Binomial Plot")
  plt.xlabel("x values")
  plt.ylabel("probability")

  return plt.show()
```

In [ ]:
```
binomplot(10,0.6)
```



Lab 9.1.1

In [ ]:
```
from scipy import stats
stats.uniform.cdf(15,5,15-5)
```

Out[ ]:
```
1.0
```

Lab 9.2.1

In [ ]:
```
from scipy import stats
stats.norm.cdf(1.5,0,1)
```

Out[ ]:
```
0.9331927987311419
```

Lab 9.2.2

In [ ]:
```
from scipy import stats
stats.norm.cdf(27726,27191,7126)
```

Out[ ]:
```
0.5299233487082249
```

Interpretation: Probability that the average cost is less than `$27726` is 0.5299233487082249

$P(X < 27726) = 0.5299233487082249$

Lab 9.2.3

In [ ]:
```
from scipy import stats
stats.norm.ppf(.75,27191,7126)
```

Out[ ]:
```
31997.413959897276
```

Interpretation: 75 percent of the average college costs are less than or equal to 31997.413959897276

Lab 9.3.1

In [ ]:
```
from scipy import stats
stats.norm.cdf(6,5,1/(10**.5))
```

Out[ ]:
```
0.9992172988709987
```

Interpretation: The probabilty that the mean length is less than 6 is 0.9992172988709987. That is $P(\bar{x} < 6) = 0.9992172988709987$

In [ ]:
```
from scipy import stats
stats.norm.cdf(6,5,1/(10**.5)) - stats.norm.cdf(5,5,1/(10**.5))
```

Out[ ]:
```
0.49921729887099875
```

Interpretation: The probability that the average length is between 5 and 6 is 0.49921729887099875. That is, $P(5 < \bar{x} < 6) = 0.49921729887099875$

Lab 10.1.1

In [ ]:
```
# Confidence interval, given confidence level C, std, n
def conf_z_mean(C,mean,std,n):
     from scipy import stats
     ebm = (std/n**.5)*stats.norm.ppf(C+(1-C)/2,0,1)
     return print("(", mean-ebm, ",", mean+ebm, ')')
```

In [ ]:
```
conf_z_mean(.90,80,15,50)
```

```
( 76.51073853896997 , 83.48926146103003 )
```

conf_z_mean(.90,80,15,50) Interpretation: The 90% confidence interval is.

Lab 10.2.1

In [ ]:
```
# marigin of error given the level of confidence C, mean, std and n
def error_t_mean(C,std,n):
    from scipy import stats
    ebm = (std/n**.5)*stats.t.ppf(C+(1-C)/2,n-1,0,1)
    return print("Margin of Error is",ebm)
```

In [ ]:
```
error_t_mean(.9,7,25)
```

```
Margin of Error is 2.3952349118731986
```

In [ ]:
```
# confidence interval given level of confidence C and x

def conf_t_mean(C,mean,std,n):
    from scipy import stats
    ebm = (std/n**.5)*stats.t.ppf(C+(1-C)/2,n-1,0,1)
    print ('mean is',mean,'and std is',std)
    return print("(", mean-ebm, ",", mean+ebm, ')' )
```

In [ ]:
```
conf_t_mean(.9,85,7,25)
```

```
mean is 85 and std is 7
( 82.6047650881268 , 87.3952349118732 )
```

Lab 10.3.1

In [ ]:
```
# confidence interval and margin of error given p and n

def conf_prop_stats(C,p,n):
  from scipy import stats
  eb1 = ((p*(1-p)/n)**.5)*stats.norm.ppf(C+(1-C)/2,0,1)
  return print(" Sample Proportion =",p,"\n Margine of Error = ", eb1,
               "\n Confidence Interval","(",p - eb1,",",p + eb1,")")
```

In [ ]:
```
p=51/120
```

In [ ]:
```
conf_prop(.9,p,120)
```

```
Sample Proportion = 0.425
Margine of Error =  0.07422753204362957
Confidence Interval ( 0.3507724679563704 , 0.4992275320436296 )
```

Lab 10.3.2

In [ ]:
```
# confidence interval and margin of error given p and n

def conf_prop(C,p,n):
  from scipy import stats
  eb1 = ((p*(1-p)/n)**.5)*stats.norm.ppf(C+(1-C)/2,0,1)
```

```
  return print(" Sample Proportion =",p,"\n Margine of Error = ", eb1,
               "\n Confidence Interval","(",p - eb1,",",p + eb1,")")
```

In [ ]:
```
conf_prop(.95,.5,120)
```

```
Sample Proportion = 0.5
Margine of Error =  0.08945970718585784
Confidence Interval ( 0.41054029281414217 , 0.5894597071858578 )
```

Lab 10.3.3

In [ ]:
```
def samp_pro(C,p,E):
  from scipy import stats
  sam = p*(1-p)*stats.norm.ppf(C+(1-C)/2,0,1)**2/E**2
  return print("sample size = ", sam)
```

In [ ]:
```
samp_pro(.9,.55,.03)
```

```
sample size =  744.0244498762386
```

Interpretation: A sample of size 745 is needed.

Lab 11.1.1

In [ ]:
```
# z test with alternative 'smaller' or 'greater' or 'two-sided'
def z_test_mean(null_mean,sigma,sample_mean,n,alternative='two-sided'):
    from scipy import stats
    z = (sample_mean-null_mean)/(sigma/n**.5)
    if alternative == 'smaller':
        pv = stats.norm.cdf(z,0,1)
    elif alternative == 'greater':
        pv = 1-stats.norm.cdf(z,0,1)
    elif alternative == 'two-sided':
        if z< 0:
            pv = 2*stats.norm.cdf(z,0,1)
        else: pv = 2*(1-stats.norm.cdf(z,0,1))
    else:
        print("alternative option error")
    return print("z test statistic = ", z, "and p-value = ",pv)
```

In [ ]:
```
z_test_mean(2,.5,2.0,25,'smaller')
```

```
z test statistic =  0.0 and p-value =  0.5
```

Interpretation: Since the p-value 0.5 is not less than alpha, fail to reject the null hypothesis. There is not enough evidence to support the offcial's claim.

---

lab 11.2.1

In [ ]:
```
import numpy as np

x=[2.5, 3.0, 1.5, 2.4, 2.6, 2.1, 1.8, 1.5, 1.7, 1.9, 3.2, 2.3, 2.0, 1.8, 1.2,
   1.5, 2.3, 2.5, 1.8, 2.2, 3.3, 2.3, 1.6, 2.5, 2.2]

mean = np.mean(x)
std = np.std(x,ddof=1)
```

In [ ]:
```
t_test_mean(1.5,mean,std,25,'two-sided')
```

```
t test statistic =  6.006870493491006  p-value =  3.3502350138547854e-06
```

Interpretation: Since the p-value is 3.3502350138547854e-06 less than alpha, reject the null hypothesis. There is enough evidence to not support the official's claim.

Lab 12.3.1

In [ ]:
```
def z_test_prop(null_prop,x,n,alternative):  #altenative can only be "smaller"
                                               # or "greater" or "two-sided"
    from scipy import stats
    p = x/n
    z = (p-null_prop)/(null_prop*(1-null_prop)/n)**.5
    if alternative == 'smaller':
        pv = stats.norm.cdf(z,0,1)
    elif alternative == 'greater':
        pv = 1-stats.norm.cdf(z,0,1)
    elif alternative == 'two-sided':
        if z< 0:
            pv = 2*stats.norm.cdf(z,0,1)
        else: pv = 2*(1-stats.norm.cdf(z,0,1))
    else:
        print("altrnative option syntax error")
    return print("z test statistic = ", z, " p-value = ",pv)
```

In [ ]:
```
z_test_prop(.5,510,1000,'greater')
```

```
z test statistic =  0.6324555320336764  p-value =  0.2635446284327688
```

Interpretation: The p-value 0.2635446284327688 is NOT less than 5%. Do not reject the null hypothesis. There is no enough evidence to support the president's claim.

Lab 12.1.1

In [ ]:
```
import numpy as np

x=[1,0,3,5,14,7,4,16,9,10]
y=[17000,20000,15000,10000,5000,9000,10000,2000,8000,9000]
```

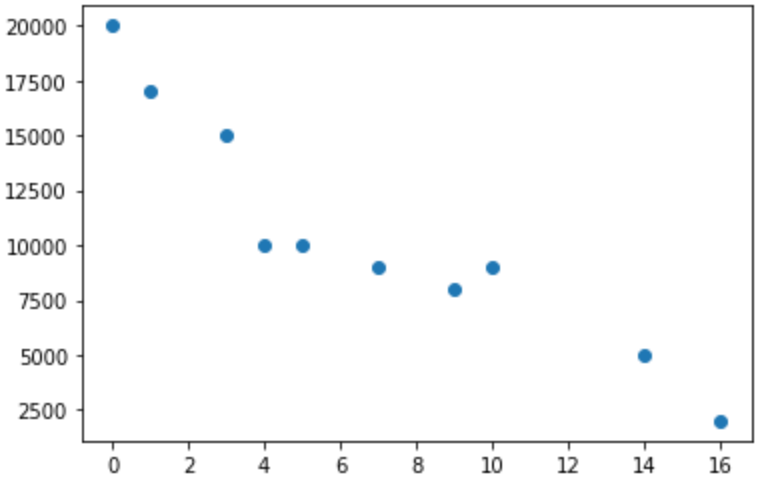
```
np.corrcoef(x,y)
```

Out[ ]:
```
array([[ 1.        , -0.93825489],
       [-0.93825489,  1.        ]])
```

Interpretation: The correlation coefficient is -0.93825489

In [ ]:
```
import matplotlib.pyplot as plt

x=[1,0,3,5,14,7,4,16,9,10]
y=[17000,20000,15000,10000,5000,9000,10000,2000,8000,9000]

plt.scatter(x,y)
```

Out[ ]:
```
<matplotlib.collections.PathCollection at 0x7fa3d949c190>
```



Lab 12.3.1

In [ ]:
```
from scipy import stats

x=[1,0,3,5,14,7,4,16,9,10]
y=[17000,20000,15000,10000,5000,9000,10000,2000,8000,9000]

stats.linregress(x,y)
```

Out[ ]:
```
LinregressResult(slope=-955.6247567146751, intercept=17093.81082133126, rvalue=-0.
9382548923679298, pvalue=5.899842619735838e-05, stderr=124.57425008729852)
```

Interpretation: slope=-955.6247567146751, intercept=17093.81082133126, rvalue=-0.9382548923679298, pvalue=5.899842619735838e-05

Lab 12.3.1 (Alternative Solution Using Array)

In [ ]:
```
from scipy import stats

x=[1,0,3,5,14,7,4,16,9,10]
```

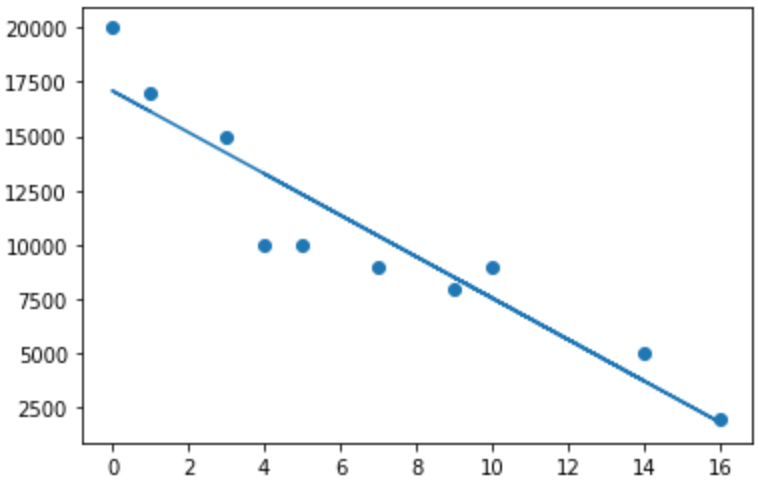
```
y=[17000,20000,15000,10000,5000,9000,10000,2000,8000,9000]

results = stats.linregress(x,y) # save results as an arry named results

print('slope =', results[0]) # results[0] is the slope
print('intercept =', results[1]) # results[1] is the intercept
print('r =', results[2]) # results[2] is the r correlation-coefficient
print('p-value =', results[3]) # results[3] is p-value for testing pop. cor. co
```

```
slope = -955.6247567146751
intercept = 17093.81082133126
r = -0.9382548923679298
p-value = 5.899842619735838e-05
```

Interpretation: slope=-955.6247567146751, intercept=17093.81082133126, rvalue=-0.9382548923679298, pvalue=5.899842619735838e-05

In [ ]:
```
import matplotlib.pyplot as plt
import numpy as np


x=[1,0,3,5,14,7,4,16,9,10]
y=[17000,20000,15000,10000,5000,9000,10000,2000,8000,9000]

x = np.array(x)
y = np.array(y)

y_hat = -955.6247567146751*x+17093.81082133126

plt.scatter(x,y)
plt.plot(x,y_hat)
```

Out[ ]: `[<matplotlib.lines.Line2D at 0x7fa3d9247f90>]`



Lab 13.1.1

In [ ]:
```
import numpy as np
from scipy import stats
```

```
obs = [306,232]
sum = np.sum(obs)
exp = [.513*sum,.469*sum]

stats.chisquare(obs,exp)
```

Out[ ]:
```
Power_divergenceResult(statistic=4.89897772395255, pvalue=0.026872598995409867)
```

Interpretation: Since the p-value 0.026872598995409867 is less than $\alpha = .10$, reject the null-hypothesis. There is enough evidence to reject the official's claim.

Lab 13.2.1

In [ ]:
```
def chi2_test_of_indp(obs):
    from scipy import stats
    chi2, p, dof, ex = stats.chi2_contingency(obs)

    print("chi-square test statitic:",chi2)
    print("p-value:", p)
```

In [ ]:
```
r1 = [1734,475,1281]
r2 = [2071,373,1938]
r3 = [678,103,935]
r4 = [534,92,560]
r5 = [5644,1661,2613]
r6 = [4094,927,1277]

obs = [r1,r2,r3,r4,r5,r6]

chi2_test_of_indp(obs)
```

```
chi-square test statitic: 1509.7409528009978
p-value: 0.0
```

Interpretation: Since the p-value=0.0 is less than the level of significance, reject the null hypothesis. That is, there is enough evidence that race and crime-class are NOT independent.

Lab 14.1.1

In [ ]:
```
import pandas as pd

x = ['male','female','female','male','male']
x_pd = pd.DataFrame({'gender':x})
print(x_pd)
```

```
  gender
0   male
1 female
2 female
3   male
4   male
```

Lab 14.1.2

In [ ]:
```
import pandas as pd
x=[20, 21, 25, 20, 29, 25, 22, 25, 21, 25]
x_df=pd.DataFrame({'data':x})
x_df.value_counts(sort=False)
```

Out[ ]:
```
data
20      2
21      2
22      1
25      4
29      1
dtype: int64
```

data	freq
20	2
21	2
22	1
25	4
29	1

Lab 14.1.3

In [ ]:
```
import pandas as pd
x=[20, 21, 25, 20, 29, 25, 22, 25, 21, 25]
x_df=pd.DataFrame({'data':x})
x_df.value_counts(sort=False,normalize=True)
```

Out[ ]:
```
data
20      0.2
21      0.2
22      0.1
25      0.4
29      0.1
dtype: float64
```

Lab 14.1.4

In [ ]:
```
import pandas as pd
x=['NO', 'NO', 'YES', 'YES', 'NO', 'YES', 'NO', 'NO', 'YES', 'NO', 'YES', 'NO',
   'NO', 'YES', 'YES', 'YES', 'YES', 'NO', 'YES', 'YES']

x_df = pd.DataFrame({'vote':x})
x_df.value_counts(normalize=True)
```

Out[ ]:
```
vote
YES     0.55
NO      0.45
dtype: float64
```

Lab 14.1.5

In [ ]:
```
import pandas as pd
x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
     72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
     82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
     88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
     103764, 105420, 109620, 146868]

x_df = pd.DataFrame({'data':x})

x_df.describe()
```

Out[ ]:

	data
count	54.000000
mean	78460.000000
std	18253.562584
min	46776.000000
25%	64377.000000
50%	78366.000000
75%	88272.000000
max	146868.000000

Lab 14.1.6

In [ ]:
```
import pandas as pd
x = [46776, 53736, 53736, 54840, 57408, 57408, 57408, 57408, 57408, 57408,
     60108, 60108, 62712, 63552, 66852, 66852, 69732, 70032, 70044, 70272,
```

```
    72024, 73380, 73380, 73380, 75408, 75468, 76248, 80484, 80484, 80484,
    82476, 82788, 82836, 83676, 84324, 84324, 84324, 85848, 87564, 88272,
    88272, 90828, 92004, 92304, 92520, 96060, 96096, 100680, 100716, 100716,
    103764, 105420, 109620, 146868]

x_df = pd.DataFrame({'salary':x})
print('skewness index is ',x_df.skew())
print('kurtosis index is',x_df.kurtosis())
```

```
skewness index is  salary    0.924462
dtype: float64
kurtosis index is salary    2.250167
dtype: float64
```

Lab 14.2.1

In [ ]:
```
import pandas as pd

url = 'https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/mas

x_df = pd.read_csv(url)

x_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   valid_import       120 non-null    bool
 1   item               120 non-null    object
 2   importer_id        120 non-null    int64
 3   exporter_id        120 non-null    int64
 4   country_of_origin  120 non-null    object
 5   declared_quantity  120 non-null    int64
 6   declared_cost      120 non-null    float64
 7   mode_of_transport  120 non-null    object
 8   route              120 non-null    object
 9   date_of_departure  120 non-null    object
 10  date_of_arrival    120 non-null    object
 11  declared_weight    120 non-null    float64
 12  actual_weight      120 non-null    float64
 13  days_in_transit    120 non-null    float64
dtypes: bool(1), float64(4), int64(3), object(6)
memory usage: 12.4+ KB
```

Interpretation: for x_df

1. class type is Pandas Dataframe
2. 120 entries from 0 to 199
3. 14 Columns
4. Info for each Column listed

Lab 14.3.1

In [ ]:
```
import pandas as pd

url = 'https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/mas

x_df = pd.read_csv(url)

x_df.describe()
```

Out[ ]:

	importer_id	exporter_id	declared_quantity	declared_cost	declared_weight	actual_v
count	120.0	120.0	120.000000	120.000000	120.000000	120.0
mean	111.0	222.0	127.458333	6743.649881	1264.702934	1306.4
std	0.0	0.0	14.641311	2991.797050	633.149971	656.9
min	111.0	222.0	100.000000	1441.012419	18.459509	19.2
25%	111.0	222.0	115.750000	4442.903914	820.314400	841.7
50%	111.0	222.0	131.500000	6010.218745	1255.597743	1305.7
75%	111.0	222.0	139.000000	8887.095370	1711.314045	1763.6
max	111.0	222.0	149.000000	14281.325362	2806.338955	2918.6

Interpretatoin: The descriptive statistics for `importer_id` and `exporter_id` are meaningless and should be ignored.

Lab 14.3.2

In [ ]:
```
import pandas as pd

url = 'https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/mas

x_df = pd.read_csv(url)

x_df.groupby(['country_of_origin']).agg({'actual_weight':['mean','std','size']})
```

Out[ ]:

	actual_weight		
	mean	std	size
country_of_origin			
China	1307.147246	627.380399	24
France	1218.459074	728.100019	24
India	1304.722531	493.478451	24
Italy	1190.657039	711.818853	24
USA	1511.163138	702.116897	24

Lab 14.3.3

In [ ]:

```
import pandas as pd

url = 'https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/mas

x_df = pd.read_csv(url)

x_df.groupby(['country_of_origin','route']).agg({'actual_weight':['mean','std','siz
```

Out[ ]:

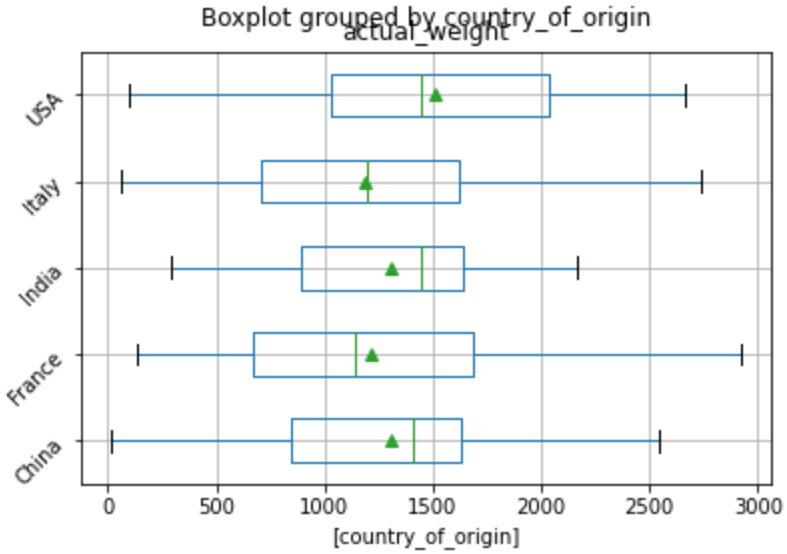
		actual_weight		
		mean	std	size
country_of_origin	route			
China	america	1281.081635	873.244407	6
	asia	1492.969412	577.826149	6
	europe	1155.582715	465.171493	6
	panama	1298.955222	654.115536	6
France	america	1142.095755	686.692379	6
	asia	1218.311778	936.931509	6
	europe	1062.014711	756.490150	6
	panama	1451.414052	644.548543	6
India	america	1007.777832	495.847993	6
	asia	1697.493097	254.063822	6
	europe	1379.484100	516.208468	6
	panama	1134.135095	459.075799	6
Italy	america	1035.376749	923.534914	6
	asia	1512.269050	974.452899	6
	europe	1002.642317	461.640254	6
	panama	1212.340040	345.152024	6
USA	america	1352.550563	895.589850	6
	asia	1730.803390	725.189852	6
	europe	1445.528489	666.483253	6
	panama	1515.770109	634.190270	6

Lab 14.3.4

In [ ]:

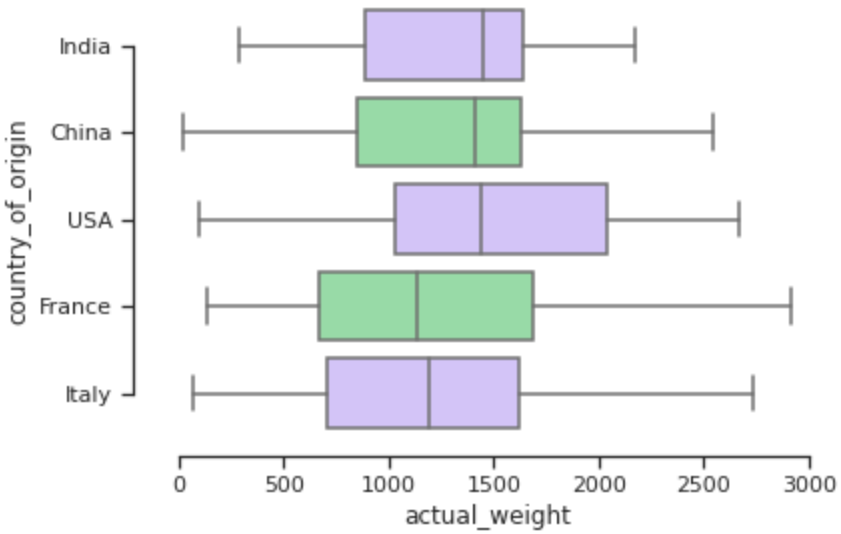
```
import pandas as pd

url = 'https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/mas

x_df = pd.read_csv(url)

x_df.boxplot(column=['actual_weight'],by=['country_of_origin'],showmeans=True,vert=
```

Out[ ]: <matplotlib.axes._subplots.AxesSubplot at 0x7f98bff49a90>



Lab 14.3.4 Alternative Solution

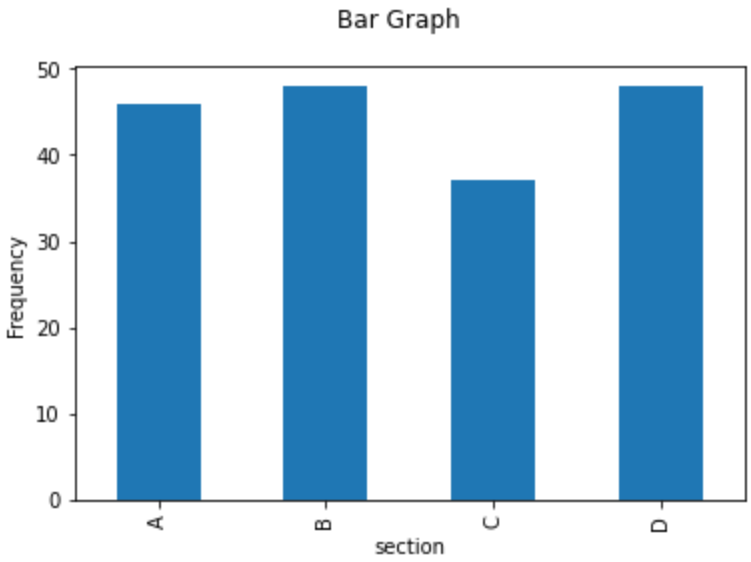
In [2]:
```
import pandas as pd
import seaborn as sns
sns.set_theme(style="ticks", palette="pastel")

# Load the example tips dataset
url = 'https://raw.githubusercontent.com/swaathi/cleaning-datasets-using-pandas/mas

x_df = pd.read_csv(url)

sns.boxplot(x="actual_weight", y="country_of_origin", hue_order="smoker", palette=[
sns.despine(offset=10, trim=True)
```



Lab 14.4.1

In [ ]:
```
import pandas as pd

url='https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv'

x_df = pd.read_csv(url)
x_df.groupby(['section']).size().plot(kind='bar')

import matplotlib.pyplot as plt

plt.suptitle("Bar Graph")
plt.ylabel("Frequency")
```

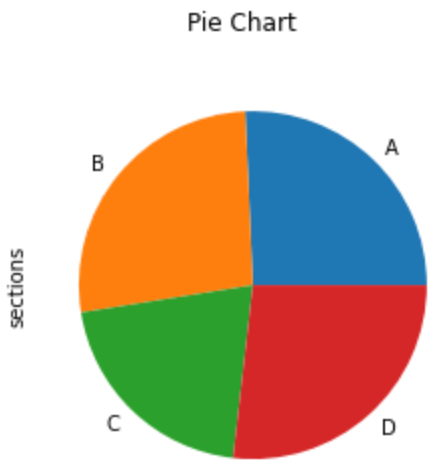
Out[ ]:
```
Text(0, 0.5, 'Frequency')
```



Lab 14.4.2

In [ ]:
```
import pandas as pd

url='https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv'

x_df = pd.read_csv(url)
x_df.groupby(['section']).size().plot(kind='pie')

import matplotlib.pyplot as plt

plt.suptitle("Pie Chart")
plt.ylabel("sections")
```

Out[ ]:
```
Text(0, 0.5, 'sections')
```



Lab 14.5.1

In [ ]:
```
import pandas as pd

url = 'https://raw.githubusercontent.com/Shreyas3108/house-price-prediction/master/

x_df = pd.read_csv(url)
```

In [ ]:
```
import statsmodels.formula.api as smf
model = smf.ols(formula='price ~ bedrooms + sqft_basement + sqft_living', data=x_df
results = model.fit()

print(results.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.508
Model:                            OLS   Adj. R-squared:                  0.508
Method:                 Least Squares   F-statistic:                     7426.
Date:                Wed, 11 Aug 2021   Prob (F-statistic):               0.00
Time:                        13:38:08   Log-Likelihood:            -2.9995e+05
No. Observations:               21613   AIC:                         5.999e+05
Df Residuals:                   21609   BIC:                         5.999e+05
Df Model:                           3
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept      8.547e+04   6673.321     12.807      0.000    7.24e+04    9.85e+04
bedrooms      -5.806e+04   2312.155    -25.111      0.000   -6.26e+04   -5.35e+04
sqft_basement    26.6854      4.409      6.053      0.000      18.044      35.327
sqft_living     308.9346      2.478    124.668      0.000     304.077     313.792
==============================================================================
Omnibus:                    14369.246   Durbin-Watson:                   1.985
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           489010.614
Skew:                           2.718   Prob(JB):                         0.00
Kurtosis:                      25.660   Cond. No.                     9.07e+03
==============================================================================

Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
[2] The condition number is large, 9.07e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Interpretation:

Interpretation:

1. See `x_df.head()` above
2. $\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$ where
   $\hat{y} = 8.547e04 - 5.806e04 b_1 + 26.6854 b_2 + 309.9346 b_3$
3. The F-statistic is 7426 with p-value of Prob (F-statistic) 0.00. Thus, the model has a significant contiribution to the prediction of price.
4. The coefficient of determination (R-squared:) is 0.508.
5. The p-value (P>|t|) for the `sqft_living` is 0.000 which is less than $\alpha = 0.05$. It is significant.

Lab 14.5.2

In [ ]:
```
import pandas as pd
df = pd.read_csv('https://raw.githubusercontent.com/simeonaman/demo/master/heart.cs
df.head()
```

Out[ ]:

	row.names	sbp	tobacco	ldl	adiposity	famhist	typea	obesity	alcohol	age	chd
0	1	160	12.00	5.73	23.11	Present	49	25.30	97.20	52	1
1	2	144	0.01	4.41	28.61	Absent	55	28.87	2.06	63	1
2	3	118	0.08	3.48	32.28	Present	52	29.14	3.81	46	0
3	4	170	7.50	6.41	38.03	Present	51	31.99	24.26	58	1
4	5	134	13.60	3.50	27.78	Present	60	25.99	57.34	49	1

In [ ]:
```
import statsmodels.formula.api as smf
model = smf.logit(formula='chd ~ ldl + age + sbp', data=df)
results = model.fit()

print(results.summary())
```

```
Optimization terminated successfully.
         Current function value: 0.553757
         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:                    chd   No. Observations:                  462
Model:                          Logit   Df Residuals:                      458
Method:                           MLE   Df Model:                            3
Date:                Wed, 11 Aug 2021   Pseudo R-squ.:                  0.1416
Time:                        14:09:16   Log-Likelihood:                -255.84
converged:                       True   LL-Null:                       -298.05
Covariance Type:            nonrobust   LLR p-value:                 3.427e-18
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -4.7372      0.768     -6.170      0.000      -6.242      -3.233
ldl            0.1863      0.053      3.487      0.000       0.082       0.291
age            0.0559      0.009      6.036      0.000       0.038       0.074
sbp            0.0048      0.005      0.898      0.369      -0.006       0.015
==============================================================================
```

Interpretation:

1. $\hat{p} = \frac{1}{1+e^{-4.7372+0.1863ldl+0.0559age+0.0048sbp}}$
2. Since LLR p-value 3.427e-18 is less than level of significance, the model contributes significantly to the prediction of `chd`.

Lab 14.6.1

In [ ]:
```
import pandas as pd

url='https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv'

x_df = pd.read_csv(url)
```

In [ ]:
```
x_df.groupby(['level']).agg({'test3_score': ['mean', 'std','size']})
```

Out[ ]:

	test3_score		
	mean	std	size
level			
high	53.773333	3.184170	60
low	53.750000	2.329727	60
medium	53.831667	2.536546	60

In [ ]:
```
import statsmodels.api as sm
from statsmodels.formula.api import ols
model = ols('test3_score ~ C(level)',data = x_df).fit()
table = sm.stats.anova_lm(model,type=1) # type 2 for equal or non-equal sample size
print(table)
```

```
               df       sum_sq   mean_sq         F    PR(>F)
C(level)    2.0     0.212333  0.106167  0.014477  0.985629
Residual  177.0  1298.037167  7.333543       NaN       NaN
```

Interpretation: Since the p-value 0.985629 is not less than α = 0.10, we fail to reject the null hypothesis that the averages for all three levels are equal. That is, there is no significant difference between the averages of `test3_score` of the three levels.

Lab 14.6.2

In [ ]:
```
import pandas as pd

url='https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv'

x_df = pd.read_csv(url)
```

In [ ]:
```
x_df.groupby(['level','gender']).agg({'test3_score': ['mean', 'std','size']})
```

Out[ ]:

		test3_score		
		mean	std	size
level	gender			
high	female	54.036000	3.487463	25
	male	53.585714	2.986946	35
low	female	53.857576	2.521412	33
	male	53.618519	2.111696	27
medium	female	53.604167	2.688863	24
	male	53.983333	2.456769	36

In [ ]:
```
import statsmodels.api as sm
from statsmodels.formula.api import ols
model = ols('test3_score ~ C(level) + C(gender)',data = x_df).fit()
table = sm.stats.anova_lm(model,type=1) # type 2 for non-equal sample sizes
print(table)
```

```
               df       sum_sq   mean_sq         F    PR(>F)
C(level)      2.0     0.212333  0.106167  0.014401  0.985704
C(gender)     1.0     0.494704  0.494704  0.067102  0.795906
Residual    176.0  1297.542462  7.372400       NaN       NaN
```

Interpretation: There is no significant difference of the `test3_score` means between the three levles afte blocking by gender. No significant difference in the average test1_scores between the two genders.

Lab 14.6.3

In [ ]:
```
import pandas as pd

url='https://raw.githubusercontent.com/simeonaman/demo/master/my_csv_file.csv'

x_df = pd.read_csv(url)
x_df.head()
```

Out[ ]:

	id	gender	section	time	level	test1_score	test2_score	test3_score
0	1	female	C	17.3	low	65.4	84.3	51.9
1	2	female	D	5.1	low	72.4	86.1	52.0
2	3	female	A	2.4	low	73.8	84.2	53.1
3	4	female	A	8.8	low	70.2	87.9	52.0
4	5	male	B	15.0	low	60.3	81.4	55.5

In [ ]:
```
x_df.groupby(['level','section']).agg({'test3_score': ['mean', 'std','size']})
```

Out[ ]:

		test3_score		
		mean	std	size
level	**section**			
high	**A**	54.207143	3.816657	14
	B	52.538462	2.417484	13
	C	54.146667	3.044636	15
	D	54.105882	3.362192	17
low	**A**	54.000000	1.942384	15
	B	53.618750	1.880858	16
	C	53.750000	2.469548	16
	D	53.623077	3.181235	13
medium	**A**	53.982353	2.444441	17
	B	53.126316	1.659264	19
	C	53.250000	2.962938	6
	D	54.627778	3.136841	18

In [ ]:
```
from statsmodels.formula.api import ols
model = ols('test3_score ~ C(level)*C(section)',data = x_df).fit()
table = sm.stats.anova_lm(model,type=2) # type 2 for non-equal sample sizes
print(table)
```

```
                      df       sum_sq    mean_sq         F    PR(>F)
C(level)             2.0     0.200742   0.100371  0.013460  0.986631
C(section)           3.0    31.217036  10.405679  1.395460  0.246013
C(level):C(section)  6.0    19.884361   3.314060  0.444434  0.848150
Residual           167.0  1245.286911   7.456808       NaN       NaN
```

Interpretation: There is no significant interaction effect between `level` and `section` of the average `test2_score`. 0.848150 > 0.10

References

Arithmetic Operators

Name	Operator	Example	Result
Addition	`+`	`5+6`	`11`
Subtraction	`-`	5-6	`-1`
Multiplication	`*`	`5*6`	`30`
Division	`/`	`5/6`	`0.83333`
Exponentiation	`**`	`5**2`	5^2 = `25`

Descriptive Stats

```
import numpy as np
import pandas as pd

x=[x1,x2,x3,...]

x_df=pd.DataFrame({'column_name':x})
```

procedure	numpy	pandas
frequency counts	`np.unique(x,return_counts=True)`	`x_df.value_counts()`
size	`np.size(x)`	`x_df.count()`
min	`np.amin(x)`	`x_df.min()`
max	`np.amax(x)`	`x_df.max()`
mean	`np.mean(x)`	`x_df.mean()`
mode	NA	`x_df.mode()`
median	`np.median(x)`	`x_df.median()`
sample variance	`np.var(x,ddf=1)`	`x_df.var()`
pop variance	`np.var(x)`	`x_df.var(ddof=0)`
sample std	`np.std(x,ddof=1)`	`x_df.std()`
pop std	`np.std(x)`	`x_df.std(ddof=0)`
weighted mean	`np.average(x,weights=freq)`	NA

procedure	numpy	pandas
weighted sample std	`(np.cov(x,fweights=freq,dd0f=1))**(1/2)`	NA
weighted pop std	`(np.cov(x,fweights=freq))**(1/2)`	NA
percentiles	`np.quantile(x,p)`	`x_df.quantile(p)`
summary stats	NA	`x_df.describe()`
factorial, n!	`np.math.factorial(n)`	NA
n random integers from a to b	`np.random.randint(a,b+1,size=n)`	NA
weighted mean	`np.average(x,weights=y)`	NA
Weighted std	`(np.cov(x,aweights=y,ddof=0)**(1/2)`	NA
correlation coef.	`np.corrcoef(x,y)`	`x_df.corr()`

NA = Not Available

Summary statistics by grouping with categorical columns:

```
x_df.groupby(['level','gender']).agg({'test1_score':
                  ['mean','std','size']})
```

Graphs

```
import pandas as pd
import mathplotlib.pyplot as plt

x = [x1, x2, x3, ...]
x_df = pd.DataFrame({'column_name',x})

CHOOSE A CODE FROM BELOW (Bar, Pie, Histogram etc)

plt.suptitle("Plot Title")
plt.xlabel("x-axis label")
plt.ylabel("y-axis label")
```

1. Bar graph (raw data)
   `x_df.groupby(['column_name']).size().plot(kind='bar')`
2. Pie Chart (raw data) `x_df.groupby(['column_name']).size().plot(kind='pie', autopct='%1.1f%%')`
3. Bar graph (frequency table) `x_df.plot.bar(x='column_name',y='frequency')`
4. Pie chart (frequency table)

```
x_df.set_index('column_name',inplace=True)
x_df.plot.pie(y='frequency')
```

5. Side by Side Bar (frequency table)

```
x_df.set_index('column_name',inplace=True)
x_df.plot.bar()
```

6. Histogram

```
freq,bins,patches = plt.hist(x,bins=5,ec='k',range=
[min(x),max(x)+1])
plt.xticks(bins)   # df['column_name']=x
```

7. Scater Plot (x, y list given) `plt.scatter(x,y)`

Linear Regression

```
from scipy import stats
stats.linregress(x_list, y_list)
```

Multiple Linear Regression

```
import statsmodels.formula.api as smf
model = smf.ols(formula='y ~ x1 + x2 +...+ xk', data=x_df)
results = model.fit()

print(results.summary())
```

Probability Distributions

```
from scipy import stats
```

procedure	scipy.stats
Binomial pdf $P(X = x)$	`stats.binom.pmf(x,n,p)`
Binomial cdf $P(X \leq x)$	`stats.binom.cdf(x,n,p)`
Normal cdf $P(X \leq x)$	`stats.norm.cdf(x,mean,std)`
Inverse Normal or p-th Percentile	`stats.norm.ppf(p,mean,std)`
t distribution cdf $P(X \leq x)$	`stats.t.cdf(t,df,0,1)`
Inverse *t* or p-th Percentile	`stats.t.ppf(p,df,0,1)`

procedure	scipy.stats
chi-square cdf $P(\chi^2 \leq ch2)$	`stats.chi2.cdf(ch2,k-1)`
Inverse chi-square or p-th Percentile	`stats.chi2.ppf(p,k-1)`

Margin of Error for Confidence Interval

```
from scipy import stats
```

Mean μ with z distribution with confidence level `C`, (population) standard deviation `std` and sample size `n`:

```
E = (std/n**.5)*stats.norm.ppf(C+(1-C)/2,0,1)
```

Mean μ with t distribution with confidence level `C`, sample standard deviation `std` and sample size `n`:

```
E = (std/n**.5)*stats.t.ppf(C+(1-C)/2,0,1)
```

Proportion p with z distribution with confidence level `C`, sample proportion `p` and sample size `n`:

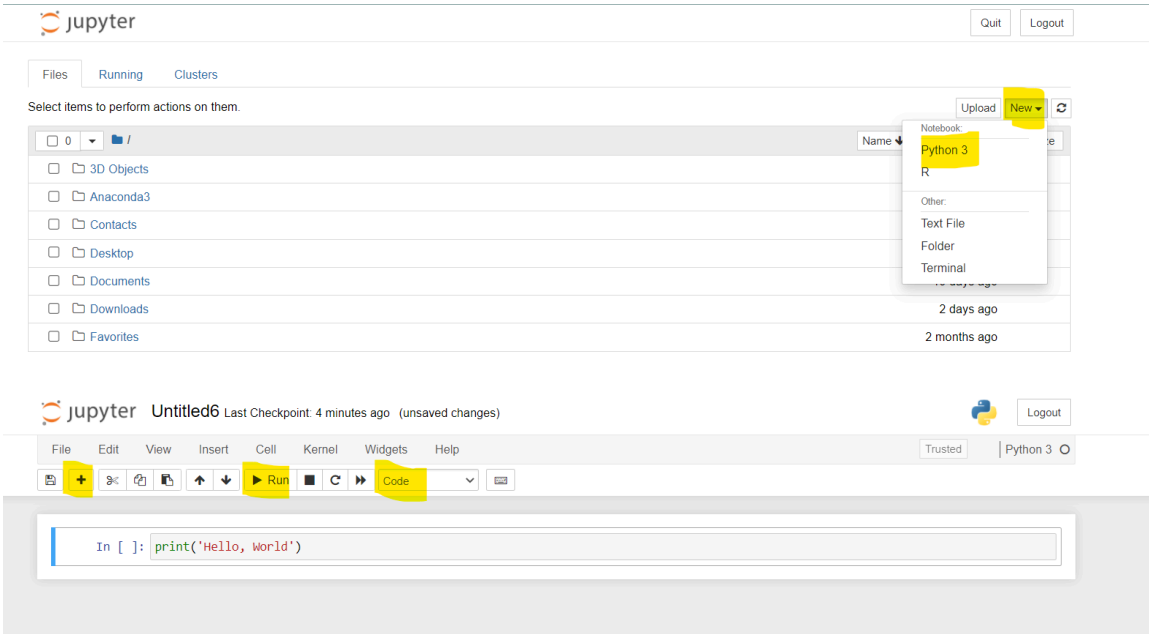
```
E = (p(p*(1-p)/n)**.5)*stats.norm.ppf(C+(1-C)/2,0,1)
```

Appendix 1

Python Computer Installation

If you prefer not to use Google Colab Notebook, you can install free Python in your computer and use the **Anaconda Jupyter Notebook**. Go to this website and select Windows, MacOS or Linux, depending on your computer. When the installation is completed, open **jupyter Notebook(anaconda3)** from the **Anaconda** folder on your computer. You can also watch YouTube videos on how to install Anaconda for Windows 10 or MacOS.

Open **Jupyter Notebook** from your Windows computer Start menu under **Anaconda** and click **new**. You can add new code-cell or text-cell by clicking + and code (see pictures below). You can run a code-cell by clicking ***run***.



Appendix 2: Random Integer Generator

Watch demo video

Appendix 3: Factorials

Watch demo video

Resources

- Python.org
- Matplotlib
- Python eBook
- Introductory Statistics eBook
- Python Tutorials
- Terminology
- Numpy
- Scipy Stats -Pandas